

Introduction and State of the Art

1 Reality-Virtuality Continuum

Occlusive VR headsets with cameras can use reality pass-through to create AR-Like experiences. AR Headsets will occlusive shielding could selectively or entirely block out reality, effectively rendering VR experiences.

2 Virtual Reality

"Everything that we know about our reality comes by way of our senses. In other words, our entire experience of reality is simply a combination of sensory information and our brains sense-making mechanisms for that information. It stands to reason then, that if you can present your senses with made-up information, your perception of reality would also change in response to it. You would be presented with a version of reality that isn't really there, but from your perspective it would be perceived as real. ... *Virtual reality entails presenting our senses with a computer generated virtual environment that we can explore in some fashion.*"

2.1 Headset/Head Mounted Display (HMD) VR

Completely supplants your visual (and mostly supplants your auditory) perception of reality. Technical properties

- Display: **Binocular 3d stereoscopic**, upto 4k+ resolution with 100Hz+ refresh rates.
- Lens: **Wide field of view** (110 degrees on horizontal typically) usually far wider than equivalent AR headsets, usually with some control of **interpupillary distance** (lens distance between eyes)
- Tracking : **Low latency head tracking, usually 6DoF** visual-inertial tracking, with tracked controllers, tracked hands all now typical
- **Variable ergonomics** (size, weight, comfort)

2.2 Smartphone-based VR

Mobile VR enclosure where the smartphone provides the guts of the hardware, used for display, rendering, etc. For example,

- Google cardboard
- Samsung Gear VR - Slightly more sophisticated enclosure (IPD adjustable lenses, integrated controls)

2.3 Inside-Out VR

- Meta Quest series, probably best headset for price/performance
- Integrates inside-out tracking for 6DoF plus hand/controller tracking, passthrough video mode for interacting with reality
- Rendering handled by a mobile chipset, so weaker than modern PC CPUs/GPUs (but still good thanks to optimizations like foveated rendering)
- Notable platform limitations (e.g. requirement to have Facebook account linked)

2.4 Outside-In PC VR

- Valve Index - Headset only works tethered to a pc
- Significantly higher fidelity as a consequence
- Tracking handles via one or more **Lighthouses** - IR beacons that are detected by the headset to allow for trilateration of position
- Downsides in terms of extend of mobility due to tether, but highest potential fidelity of rendering.

Similar to spatial projective AR, effectively just with a far greater display coverage for occluding reality - Room-Scale Projection (CAVE - Cave Automatic Virtual Environment, multi-wall environments) - Dome Projection (Hemisphere/spherical displays - less or no notable seams) - Vehicle Simulators (Simulating visual displays in windows with real physical vehicle interior)

- Virtual reality is not necessarily rendered visually or aurally
- Some devices have been created to make use of taste/smell

3 Non-HMD Virtual Reality

3.1 Projection/Large Display VR

Similar to spatial projective AR, effectively just with a far greater display coverage for occluding reality

- Room-Scale Projection (CAVE - Cave Automatic Virtual Environment, multi-wall environments)
- Dome Projection (Hemisphere/spherical displays - less or no notable seams)
- Vehicle Simulators (Simulating visual displays in windows with real physical vehicle interior)

3.2 Key Characteristics of VR

Immersion - Being surrounded by a vivid illusion of a virtual world, conveyed across one or more senses, preferably in 3D(where applicable) and rendered correct to your perspective/movements.

Interactivity, Plausibility, Autonomy - That your actions impact the virtual world realistically, and that you can interact with/react to the virtual world.

Presence - That you feel a sensation the virtual world *is* your reality (driven by immersion).

4 Augmented Reality

Reality, with some virtual elements

- Mobile / Handheld, Smartphones
- Spatial AR / Projection Mapping
- Heads-Up Displays or Overlays / HUDs
- Headset / Head-Mounted Display / Immersive AR
- Retinal / Contact Lens
- Multi-sensory displays

4.1 Mobile/Handheld AR

Pros	Cons
Low requirements Easy to develop for Pragmatic Can judge virtual elements against real backdrops	Limited View window Limited immersion/presence Limited/no 3D interaction Typically weak AR

4.2 Spatial AR/Projection Mapping

Pros	Cons
Accessible to all in space High resolution/fidelity Limited sensing requirements Immersive - exploits physical environment	Needs digital twin of the physical object Often perspective dependant Need multiple projectors to cover 3D volumes or large surfaces Requires installation process

4.3 AR-in-VR Passthrough Headsets

e.g. Meta Quest 3, Apple Vision Pro

Pros	Cons
Personal and Private augmentations of world Augmentations rendered correctly to your perspective Much more powerful tool than mobile AR Binocular, depth perception	Higher processing requirements Higher hardware requirements (field of view, rendering technology) Form factors not yet consumer friendly / fashionable / wearable all day (especially for passthrough AR) using VR headset which occlude reality by default and need cameras to show you reality expensive Typically referred to "Strong" AR - Very accurate tracking, seamless integration with reality

4.4 Retinal Displays

Pros	Cons
Discreet, wearable, high FOV	Powering the displays Communications Fidelity Comfort basically everything, super difficult

5 VR and AR Differences

VR prioritises immersion, whereas AR alters perception.

- **Extended Reality:** The virtual supplements the real, i.e. the environment contains more information.
- **Diminished Reality:** The virtual removes the real, e.g. Black Mirror
- **Altered Reality:** The virtual transforms the real, e.g. spatial AR building
- **Hybrid Reality:** The virtual completes the real, e.g. physical illusions box
- **Extended Perception:** The virtual translates perceivable aspects of the real world, such as radiation or ultrasound, or from impairments, to information we can perceive with our available senses

5.1 Use Cases of AR

The death of physical displays and the rise of spatial computing : Who needs a smartphone or laptop display when the AR headset can render virtual displays for entertainment, productivity, communications, etc. anywhere, anytime.

Augmented Cognition - Acquiring knowledge, what the brain does : Understanding and supporting cognitive state/user's mental model. Supporting and enhancing decision making, conducting tasks (from manufacturing to medical). Enhancing short and long-term memory capacity enhancing and direction attention.

Augmented (artificial?) Intelligence - using knowledge, capacity for action : How social AI becomes an ever present part of your reality. How you seamlessly communicate with others across languages. Meta Ray Ban collaboration is good indicator here, HUD glasses as an interface for Meta AI.

Augmented Perception / Perceptual Manipulation : Highlighting important things, blocking unimportant things, making the invisible visible, extending your perceptual reach, etc. **Personalising Reality :** For good, addressing social anxiety, making your environment more pleasant. For bad, altering perception of world to reinforce views, stereotypes, advertising everywhere, etc.

5.1.1 Where is consumer AR heading?

Near Term

- Fashionable HUD-type visual AR glassed (limited FOV)
- Auditory AR as standard - microphone arrays, integrated voice assistants
- Connected eyewear/smart glasses - low power, all day eyewear
- Smartphone provides bulk of the rendering power
- Some limited camera integration - e.g. capture live events in real time with object recognition, ethical challenges presented
- Pass-through VR as dominant consumer experience of head worn AR in suitable environments

Long Term True 6DoF AR optical see-through headsets/glasses. This is a long way off.

6 XR Hardware & Sensing

6.1 Examples

Meta Project Aria

- 1 x 110 degree HFOV Rolling Shutter High Resolution RGB camera, up to 8mp
- 2 x 150 HFOV/ 120 degree VFOV Global Shutter mono cameras for SLAM and hand tracking, 640 x 480 px

- 2 x 80 degree DFOV Eye-tracking Global Shutter mono cameras with IR illumination, 320x240 px
- 3 x IMU, Barometer, and Magnetometer (1 IMU is 1KHz and the others is 800Hz)
- 7 x 48 KHz spatial microphones
- GPS, Bluetooth, and Wifi

Meta Orion Glasses

- Micro LED projectors with waveguide optics
- Silicon carbide lenses for higher FoV (70 degrees)
- 98 grams - lightweight
- Several cameras in two frames
- 2hr battery life
- Interact using eye tracking, hand tracking, voice, and neural wristband (EMG)
- Cost around \$10000 (not available for consumers)

Apple Vision Pro

- Two high-resolution main cameras
- Six world facing tracking cameras
- Four eye tracking cameras
- TrueDepth camera
- LiDar scanner (time-of-flight)
- Four inertial Measurement Units (IMUs)
- Flicker sensor
- Ambient light sensor

7 Sensing

7.1 Directional / Spatial Microphones

For example, the 6-mic array in the apple vision pro using directional beamforming, meaning they can be more sensitive/selective to sounds coming from certain directions (e.g. speech from user, speech nearby etc.)

Consider what we can learn from what we can hear (from you, from others, from environment) through e.g. Social Signal Processing.

7.2 Cameras

At various wavelengths and positions depending on what we want to be able too see. we use camera to track our position in the world so we can render 3D content that is perspective correct. so that we can understand the world around us, i.e. tracking bystanders, detecting objects, rendering things that respect real world occlusion, etc. And to understand the user, i.e. hand gestures, facial gestures, torso/leg movements, eye movements, etc.

Camera placement is important, as using cameras in different places gives us different areas being tracked by the cameras, for example, meta quest 2 tracked in front of and above the user, but in quest 3 less of the space above the user is tracked but the space to either side or below the headset is tracked more. (quest 2 left, quest 3 right)

7.3 Depth

Binocular Stereo Vision emulates human vision, relying on two parallel view ports and calculates depth by estimating disparities between matching key points in left and right images. Usually passive, and no external light is required other than ambient light. Can be active (e.g. projecting textures) to help differentiate key points. Constrained by distance between lenses, e.g. Zed2 works from 2..20m

Time of flight (like in the vision pro via a LiDAR module) emits a laser beam, and detects the reflected light. The distance to an object is calculated based on the time taken for the light to return.

7.3.1 StereoVision vs ToF vs Structured Light

ToF more resilient to ambient light, and works well in dim lighting as well as outdoors. It tends to have greater accuracy, speed, and range, and is better suited towards dynamic scenes.

Structured Light more suitable for static scenes indoors (sunlight can interfere with projected light patterns) and higher accuracy in shorter ranges (face recognition, gesture recognition, etc.)

Stereo Vision Both ToF and structured light can work in a range of poorer conditions (i.e. Lack of ambient light, lack of texture). In contrast **Stereo Vision** needs ambient light, and there's more uncertainty in the depth information, particularly when there is an absence of distinguishable imagery common to both cameras (e.g. repeating/no textures). Depth is more limited, plus more processing power needed.

7.4 IMU, Barometers, Temperature

IMUs or Inertial Measurement Units measure specific force, angular rate, the orientation using a combination of accelerometers, Gyroscopes, and Magnetometer. An IMU runs at very high sample rates (800+) to give super low latency motion tracking for head/controllers, but this is noisy - error grows over time and with drift

7.5 Physiological Sensing

. Lots of options here:

- EOG - Electrooculography, e.g. non-camera based eye movements
- EEG - Electroencephalogram - Brain activity
- EMG - Electromyography - Muscle activity
- EDA - Electrodermal Activity - Skin
- PPG - Photoplethysmograph - Heart Rate Variability (HRV), Oxygen Saturation (O2), etc. as indicators of stress
- FNIRS - Functional near-infrared spectroscopy - Brain activity using IR light to examine oxygenation in brain
- Temperature (fingertip, head, etc) - Galvanic Skin Response (GSR) for sweat, etc.

7.6 GPS, GNSS, Wifi, Bluetooth, Cellular

GNSS/GPS gets world position accurate to a few meters if satellites are visible

Wifi/Bluetooth/Cellular can triangulate position relative to where the radio frequency broadcasts are coming from

7.7 I/O: Pressure, Capacitive Touch, Haptics

Mostly on peripherals such as controllers (pressure sensitive buttons, capacitive surfaces) but also on headsets, such as the sides of glasses. Can also use IMUs to detect gestures, e.g. taps on the side of glasses or pen

8 Tracking in XR

Why do we need tracking? For Perspective dependent rendering, e.g. rendering graphics based on head orientation/position movements, like turning head and/or walking around physical environment. Essential for visual perception of 3D/depth. We also need to track controllers/hands to be able to reach out and interact with virtual objects

8.1 Tracking Technologies

Active sensors requiring emission of signal to track, i.e. GPS, Wifi, cellular, magnetic, ultrasound. Typically uses transliteration approaches to determine position relative to multiple signals, to varying degrees of accuracy.

Passive sensors, retrieve measurements without interacting with environment, i.e. Inertial sensing/IMUs - high sample rate but noise. Or Optical tracking, which is more accurate but lower sample rate

Hybrid/ Sensor fusion - combining Different tracking technologies, e.g. visual inertial odometry, SLAM tracking. Most commonly in XR tech we deal with the combination of IMUs and optical sensing in various configurations.

Tracking Degrees of Freedom (DoF) The ability of an object to move around in space, or more generally, the number of independent ways by which a dynamic system can move. 0 DoF is no movement, 3 DoF is either orientation or position only, 6 DoF is orientation and position

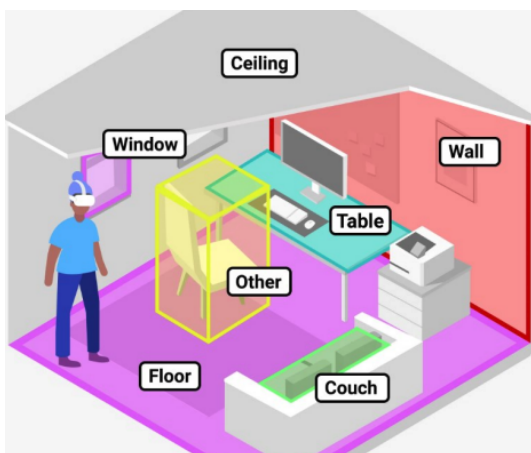
8.1.1 Outside-In vs Inside-Out Positional Tracking

Outside-In, e.g. Oculus Rift CV1, Optitrack, utilises sensing in the environment, i.e. beacons/cameras to track headset and peripherals.

Inside-Out, e.g. Meta Quest, AR headsets/smartphones, utilise Simultaneous Localization and Mapping (SLAM) to track the environment (or fiducial/passive/active markers), often as well as hands/peripherals tracked optically.

8.1.2 Scene Model

Regardless of tracking solution, we need to know where spatial anchors and world locked content are, giving reality based occlusion and **context awareness**, which is basically fundamental to spatial computing.



8.2 Pros and Cons of Tracking Solutions

Inside Out :

Pros	Cons
Markerless setups are self-contained, easy to setup, and more mobile friendly Lower cost No need for excess cables/other devices	Cameras need to be able to fix onto objects(or markers) in environment to operate Positioning is relative, not absolute (a problem from multi-user experiences)

Pros	Cons
High accuracy absolute positioning Multiple devices can use the same sensing/tracking	External hardware required, often costly, and needs to be securely mounted in environment becoming less necessary given advances in inside-out tracking

Outside In

8.3 Visual Position Systems (VPS)

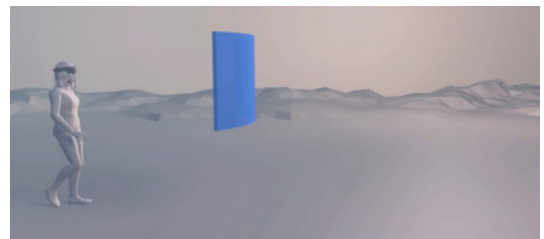
GNSS can only get us so far, and only works outdoors. How can we more precisely and robustly localise users?

VPS uses already acquired knowledge about locations, i.e. existing mapping data from streetview and the likes, and much like inside out tracking on a local scale, identifies shared landmarks to identify where the device is in relation to those landmarks

9 Spatial Registration

9.1 Egocentric perspective

Describes an objects location with respect to the perceiver's perspective.



Egocentric perspective is

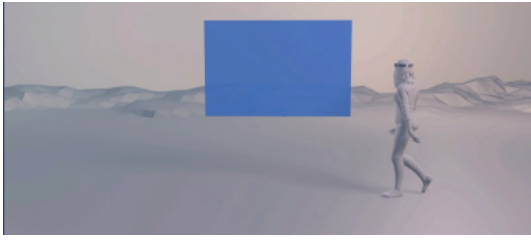
- Mobile, i.e. moves with the person
- It is reachable, the distance from user to virtual element remains constant
- It is consistent, layout remains the same regardless of low cost
- Low cost, only needs 3DoF, content floats in space relative to user.

However, there are some negatives to this:

- Bad for nausea and discomfort (implementation/context dependent)
- Bad for collaboration, as there's no shared frame of reference
- Breaks implicit association between context and content that exocentric provides
- Social acceptability - content hover in air could be hovering over a person
- Reduced accuracy of mid air touch, due to unintentional window motion

9.2 Exocentric

Describes an object's location independently of the perceivers perspective or location



Advantages of exocentric positioning are

- Memory - aids recall by placing elements in spatial context.
- Augmentation of environment, any object can have a rich display surface attached.
- Where overlaid on reachable physical surfaces, provides tactile feedback and possible additional sensing
- When overlaid on body, recall and reachability improved thanks to proprioception
- More immersive, integrating into environment.
- Allows for collaboration with shared views and activities
- Rendering to a known surface allows for content to be better rendered for clarity
- Can delineate spaces by attaching different activities and capabilities to different rooms, surfaces, or objects, helping to manage cognitive load and reduce visual distraction

However, there are some negatives to this:

- Need SLAM, depth cameras/indoor positional tracking
- Reachability impaired
- Discoverability required and augmentations may be occluded
- Content layout and managing transitions between spaces problematic
- Requires surfaces available to appropriate and comfortable depths

10 XR Design Challenges

10.1 Reality Awareness

Typically used in shared, social spaces (e.g. the home, office) with bystanders also present. These spaces were not designed for roomscale VR - they have obstacles and unpredictable bystanders.

Headsets support guardian boundaries, and more recently "space sense" on meta platforms for bystanders, but current solutions are limited

10.2 Spatial Interaction

What interaction techniques and input modalities to use? This has an impact on precision, speed, usability, and fatigue. Should consider how they adapt to the context and impairments of the user. Crucially missing haptics compared to real world interactions.

How do we design 3D UIs that are ergonomic, comfortable, usable, and safe? And how do we enact locomotion to navigate virtual worlds effectively?

10.3 Accessibility

XR devices are inherently ableist in terms of physical characteristics, range of motion available. Impairments may be

situational/environmental, temporary, or permanent, impairments can significantly impact usability of XR interaction techniques (e.g. raycast controller with broken arms)

Accessibility of XR experiences is currently woeful, support at a platform level is poor. Most XR experiences rely on assumption of people without significant disabilities, ignoring around 18% of UK

There are toolkits to support some visual accessibility tools, e.g. SeeingVR, but technology still immature. It took years for the web accessibility standards to come to fruition, similar effort now needed in XR - this is a rapidly advancing space.

10.4 Ergonomics / Comfort

Of the headsets themselves - weight, form factor, necessity of straps, comfort, etc. We are not at the point of an all-day wearable VR headset yet.

Also of the contexts and interfaces we use, for example, for productivity we've previously mentioned that virtual displays rendered by an XR headset have notable advantages over physical monitors - able to better adapt to user's posture/position

10.5 Social Acceptability & Use in Social Settings

Would you use mid air gestures and handheld controllers to, for example play beatsabre in certain scenarios? for example on a plane, in a park, etc.

10.6 Perceptual Realism & Presence

Branch of HCI largely focussed on interaction fidelity, leveraging novel haptic gloves, force feedback, controllers to create more realistic interactions with virtual content. Continual advances in other senses

10.7 Cybersickness

Multiple causes, particularly in VR. Fairly unique in HCI, if we design experiences poorly, you can actively make people feel sick. Once onset occurs, users can be nauseated for hours.

10.8 Safety and Privacy

If we experience harassment in social VR or violent/sexual media in VR, how might this impact the user? Will prolonged use of VR or AR alter or harm how we portray ourselves, others, and the world around us? Is there a potential for lasting psychological trauma?

Is there such a thing as privacy in a world where everyone is wearing AR glasses? How will this ultimately impact society?

10.9 Lack of Standards

Common across all these points is a lack of standards. Lots of competing toolkits/frameworks/platforms and third party apps are wild west. Imagine if your mouse behaved differently in every application, and every UI looked vastly different. Likelihood is that usability is impaired, which is the case for AR/VR right now.

Same applies to layout of applications in a virtual environment - lots of freedom poses a big challenge.

11 Summary

VR and AR are distinct but converging technologies, approaching mixed reality from different angles. VR prizes immersion, whereas AR alters perception.

XR devices pack lots of sophisticated sensing, understanding where they are, what the user is doing, what is happening in the world around them.

Inside-out tracking drives current 6DoF experiences, but we need tools like VPS if we are to move towards world-scale AR experiences and shared spatial anchors.

XR is tough to design apps for - a lack of standards, accessibility, ergonomics, and more. Not many computing technologies have the capability to make users feel sick.

Perceptual and Moving Through Worlds (Locomotion)

1 Interaction Modalities on HMDs

Trackable objects external to HMD - active tracking with 3DoF using IMU - controllers like those on Oculus Go, smartphones. 6DoF using IMU + Optical - tracked pens, controllers, tracker pucks (which in turn can be used to track feed/torso etc). Passive tracking - CV object recognition, marker tracking, hands.

Other interaction modalities might be gaze, speech, physical HMD controls (volume rockers, buttons), Non tracked peripherals (trackpad, mouse, keyboard). Some of these are mutually exclusive - e.g. you only have two hands, pick between hand or controller.

There are a variety of interaction design patterns, lots of innovation. This is advancing quickly, and design depends on application.

There are two common building blocks that enable the creation of most XR interactions, namely collision and raycast-ing. These are found in most gaming engines.

2 Collisions and Colliders

3D objects have a mesh (collection of vertices defining their shape), which is the 3D geometry of the object. They also have a collider, which can be quite different geometry to the underlying object for performance reasons. 3D engines such as unity provide primitive collider types, e.g. cube, sphere, capsule, mesh.

For direct touch interactions, when a collider intersects with another, it can fire events, typically onstarts, continue, stop. Either trigger events if the collisions is non-physical (e.g. knowing if a player is in an area). Could also be collision events if the collision is intended to affect physics of object.

Fundamentally, colliders need to overlap for these events to occur. this is fine for direct interactions, but what about selecting an object outwith reach?

2.1 Raycasting

The core building block of mid-air controller, pen, or gaze interactions in XR. Given an origin position, direction vector,

and a distance, the engine checks what colliders are on the path of this line or ray.

Lots of things govern whether a collision will occur, such as what layer the objects are on, e.g. you could have UI elements on a UI layer so they aren't inadvertently hit by your gun raycast but work with your UI raycast.

3 VR and Perception

The goal of VR is to create the illusion of reality. Fooling human perception by using technology to generate artificial sensation. We need to overcome the **device gap** - where we are aware that a device is mediating what we are experiencing and thus aware that the VR experience is not real.

We perceive reality through our senses - sight, hearing, touch, taste, smell, for example, but there are more. Proprioception (sense of body position), Equilibrium & Acceleration (sensed by vestibular system), Nociception (sense of pain), Temperature, etc.

There are two main processes, sensation - gathering information, and perception - interpreting information. Of our senses, vision often dominates, and is what majority of brain neurons involved with sensing process.

3.1 Challenges with Vision

3.1.1 Eye Strain

Vergence Accommodation Conflict - mismatching cues between *vergence* - inward/outward rotation of eyes to fixate on objects and perceive them as singular and *accommodation* - focusing mechanism of the eye.

Disparity between the physical surface of the screen and the focal point of the simulated world. The result of these is an unpleasant feeling and eye strain.

3.1.2 Latency

Motion-to-photon latency - If your frame rate tanks, you are increasing latency. Consider the time it takes for the tracking algorithms to take in camera and IMU data, and determine/update position and orientation, and the sample rate by which this is achieved.

Optical tracking cameras run at 60Hz, inertial tracking at 1000Hz, headset might render at 90-120Hz, so about 8-11ms between frames, or worse if you don't maintain a stable framerate.

This leads to degraded visual acuity, where the scene still moves when head stops; degraded user performance, it becomes harder to select or move objects; breaks-in-presence, undermining perceptual realism; cybersickness

3.1.3 Fidelity

Foveated displays and foveated rendering.

The fovea in your eye contains the highest concentration of photoreceptor nerves. We can save performance/power by concentrating on rendering at the highest resolution for the fovea

foveated rendering - using eye tracking to know where the user looks, and then rendering at the highest resolution there

Foveated display - combining high and lower resolution displays for central vs peripheral vision

	Human Eyes	Typical HMD
Field of View	$210^{\circ} \times 135^{\circ}$	$110^{\circ} \times 110^{\circ}$
Frame rate	1800 Hz	90 Hz
Angular Pixel Density	60–200 pixels/degree	10–13 pixels/degree
Dynamic Range	96 bit (1:10 ⁹)	24 bit
Stereo Overlap	120°	110°

3.2 Bounds of Hearing

Sounds can be localised using 3 coordinates - horizontal direction, elevation, and distance. Much less restrictive bounds compared to vision (360° horizontal) at good accuracy. Up to 20° error to side of head, largest error occur at the above/below elevations and behind head and with front/back confusion.

3.3 Bounds of Touch

Skin has various mechanoreceptors that respond to pressure, stretching, and vibration. Either slow or rapidly acting

Body Site	Threshold Distance
Finger	2–3 mm
Cheek	6 mm
Nose	7 mm
Palm	10 mm
Forehead	15 mm
Foot	20 mm
Belly	30 mm
Forearm	35 mm
Upper Arm	39 mm
Back	39 mm
Shoulder	41 mm
Thigh	42 mm
Calf	45 mm

Type 1 at surface - light discriminate touch, type 2 deep in dermis - heavy and continuous touch. Spatial resolution varies significantly.

3.3.1 Proprioception and Kinaesthesia

Proprioception is joint position sense, awareness of movement and position of body parts. Joints closer to body are more accurately sensed. Users know hand position accurate to 8cm without looking at them.

Kinaesthesia is joint movement sensing. Sensing muscle contraction or stretching, helping with force sensation.

3.4 Vision, Audio, Haptics = Multisensory Perceptual Illusion

Vision - high resolution per-eye displays and per-eye rendering to create stereo depth cues. High vertical/horizontal FoV displays.

Hearing - Head tracking to adapt visual and auditory cues based on movement through space.

Touch/haptics - Controllers with haptic feedback, aligning virtual elements with physical surfaces/objects for haptic realism.

End result is perspective correct imagery and spatial audio that adapts as we move, with sensations of touch as we interact with virtual objects. Induces a sense of presence that you are in the virtual world, and that the virtual world is plausible and real.

4 Moving through Virtual Worlds

How do we move through VR environments in a way that maintains the perceptual illusion?

4.1 Locomotion

Movement or the ability to move from one place to another

Navigation is a combination of way-finding (cognitive component) and travel (motor component - translation and rotation)

In virtual worlds as in the physical world, we may want to get from point a to point b. That target position may or may not be known to the user. We might also just want to explore the world freely.

When the virtual world is bigger than our real world environment, or goes beyond the limitations of our real world movement, we love the 1:1 mapping of real to virtual space - meaning users can't simply walk around in roomscale VR to fully navigate the virtual space. So, we need a **locomotion technique**.

4.1.1 Constraints

- Available inputs/sensing
- Physical capability and accessibility - situational/temporary/permanent impairments for example
- Physical bounds/playspace.

4.1.2 Challenges

How we get from a to b depends on what we are prioritising in our VR experience, whether its **Time/speed; Effort; Usability; Realism; Social acceptability; Immersion/presence; escapism; Excitement; Comfort/nausea**

4.2 Comfort and Nausea

Vection is the illusion of self-motion. Our body perceives our self-motion through various senses, and uses this to arrive at certainty around what movement is happening, integrating multisensory cues.

Different senses are argued to have different weightings in this process. Our brains in particular prioritise visual and vestibular input.

A proposed explanation for motion sickness is that passive movement creates a mismatch information relating to orientation and movement supplied by the visual and vestibular systems, it is this mismatch that induces nausea. This is a conflict around trained expectations (habituation) and self-motion.

4.2.1 Locomotion and Sensory Conflict

Motion sickness occurs when visual system doesn't perceive self motion that vestibular system does, e.g. reading in a moving vehicle, or physically moving in 3DoF VR

Cybersickness is when visual system perceives self motion that the vestibular system does not. For example, moving in VR and perceiving this visually, but not moving physically.

4.2.2 Cybersickness

25-40% of VR users get simulator/cybersickness. There are multiple causes for this.

- Sensory mismatch
- Low frame rate and high latency - VR experiences need to be performant
- Unstable tracking / mismatched motion - headset loses tracking or tracker reports wrong movement position

- Motion parallax / depth cues - apparent motion of object is different closer or further away
- Wide FoV - peripheral vision more sensitive to motion
- Vergence / Accommodation conflict
- Eye Separation

4.2.3 Interpupillary Distance

Refers to the distance between eyes, specifically between the centers of pupils. A paper found that some headsets were made with assumed IPD ranges based on averages for men, which consequently meant that women experienced more cybersickness.

4.2.4 Back to Sensory Conflict

Locomotion techniques that involve morevection without real motion, particularly the perception of acceleration rather than linear movement, are more problematic with respect to cybersickness. They create a bigger sensory mismatch/conflict.

We need to try to avoid this sensory conflict, or we need to train users up to be resilient to it. Place an emphasis on artificial locomotion that incorporates physical locomotion movements to minimise sensory conflict.

5 Navigation

We have many different options for locomotion techniques, with differing priorities;

- High Speed - Teleporting: User moves a predetermined distance in the looking direction by pressing a button.
- Low Effort - Sliding Joystick: Movement speed on ground proportional to joystick, up to max velocity. Direction modulated by head and joystick direction.
- *Even* Lower Effort - Gaze-Direction Steering: User moves forward with a constant speed when a button is pressed
- Comfort - Minimap: Using an in-game minimap with controllers, select a location on the map to teleport to. Could be predetermined locations to select from
- Usability - Simulating natural, physical movement: e.g. using a stationary bike pedalling and braking for forward/back motion, and handlebars for directions

There are many examples of uses of realism and motion simulation in VR, for instance, omnidirectional treadmills for infinite walking, in-car VR simulations, or hamster ball style environments.

On the other side of things, we also have immersion and escapism, for example **Birdly**, a VR experience in which users fly through the air, with gaze for directional input and buttons to flow higher or lower.

5.1 On Orientation Changes/Turning

Most locomotion techniques rely on physical orientation changes, the vr user turns their head/body to turn in VR. But, for seated VR in particular, we cant rely on this being possible. Some other common options are;

- Continuous/Free turning - using control input to turn smoothly, could be problematic from cybersickness perspective, there is a sensory mismatch.
- Discrete/snap turning - turning in chunks of x degrees at a time on the yaw axis based on input
- Rotational Gain

6 Way-finding

Goal is to build spatial knowledge, a mental model of environments so that you can determine and maintain awareness of where you are. 6DoF makes way-finding hard, we have unfamiliar, unbounded worlds, virtual navigation techniques may confuse and disorient or otherwise interfere with creating this mental model.

We support way-finding by creating landmarks, paths and breadcrumbs, using map-like topological knowledge, and procedural knowledge

We can also use maps, a World-in-miniature, to give a second viewport into the virtual world that you can also use to e.g. pick teleport destinations. Using maps gives more landmark knowledge and spatial context.

We could also use paths, a compass, or radar to help guide users.

The choice of locomotion technique has an impact on this, if you can teleport large distances, you lose the opportunity to develop spatial memory about the route taken. Could instead modify teleport so you face towards where you came, or limit teleport distances to short jumps. Similarly, if you move fast this might impair visibility of landmarks, could fix this by simply having bigger and more noticeable landmarks or limiting movement speed.

6.1 Designing Locomotion

Consider the real play area in relation to the virtual play area, consider the constraints and limitations of the users movements, the available inputs, and the desired outcomes.

Aim to provide multiple and alternate locomotion techniques to accommodate the above. Match the navigation technique to the application, target user, and impairments and constraints.

Try to avoid techniques that confuse the user and undermine their mental map of the environment, e.g. if teleporting, use projected avatars to help the user understand the path taken.

Avoid significant self motion that isnt controlled by the user in some way, e.g. if the character falls off a cliff (or similarly big event happens) move to third person view.

7 Minimising Nausea/Cybersickness

7.1 Avoiding Cybersickness

Maintain the recommended framerates. Avoid significant **vection** in seated/standing experiences. Avoid abrupt movements of content, especially the greater the display area this content consumes. Perception of visual movement conflicting with vestibular system might lead to nausea.

Be careful with use of egocentric head-locked content, this occludes the visual perception of head motion which can again lead to sensory mismatch. Is content needs to be brought to the user, move its slowly and smoothly. Use short or no animation, slow down timing.

Duringvection, use real movements instead (if you have roomscale VR). Even if its not the same, have some congruent physical self-motion, e.g. jumping on the spot, pumping arms, etc. all create some vestibular sensed movement that eases the sensory mismatch. We want to minimise the perception ofvection when it occurs, and provide users with congruent visual cues that aren't moving.

7.2 Using Movement

The reason roomscale VR is so popular, is that it feels real because the movement is real, what is perceived virtually matches what is perceived by the vestibular system. But, real movements constrain the user to the real playspace bounds - how do we get around those constraints.

We can even use different real movements for congruent self motion, congruent vestibular input, even if its not the expected input, can lead to distraction/cognitive load, and so users feel in control (habituation) - this is why drivers feel less motion sick than passengers, they are in control and can anticipate what will happen.

Peripheral vision is most sensitive to motion, so we can just block peripheral vision during vection, using fov restrictors/vignetting/tunnelling.

8 Perceptual Manipulations

Alter the human multi-sensory perceptions of our physical actions, and reactions to nudge, alter or influence the users physical movements. The VR headset does not have to create a 1:1 matching portrayal in virtuality of what happens in reality.

Visual dominance means that visual perceptual manipulators can be particularly compelling, as you innately trust your visual perception of motion. Some manipulators can may be indistinguishable from reality and imperceptible, some may be perceptible but still plausible and believable.

Perceptual manipulators for locomotion are

- Translational and rotational gain
- Redirected walking/movement
- Impossible or overlapping spaces (moving platforms, non-euclidean spaces)

8.1 Gain

Lets say we want to use real movement to enact locomotion, but the user is constrained in some way, whether it may be limited turning space or limited roomscale space. We can amplify/multiply real movements in VR with little-to-no impact on cybersickness, as a form of perceptual manipulation - **Applying gain** For position, this is *translational* gain, for orientation, *rotational*

8.1.1 Translational Gain

By applying translational gain, we effectively grow the perceivable play area.

Depending on what actions/movements people are experiencing, we can apply extreme gain with little to no impact to cybersickness. Around 1.6x is imperceptible to users, with 2x+ being noticeable but usually not problematic.

With more dynamic and interactive movements, e.g. reaching for/grabbing objects, around 1.75x-2x gain is the limit before instability occurs. Having an active task reduces how noticeable any mismatch is.

8.1.2 Rotational Gain

Detection thresholds of around 1.13x to 1.42x, roughly 20-40%, in practice, this means a 45° turn is perceived as a 56-64° turn. Detection depends on a lot of factors; rotation speed, repeated exposure, VR experience, seated vs standing, FoV of headset, etc. Secondary tasks increase imperceptible thresholds, making users less sensitive/aware.

Has Overt and tolerable thresholds, 2x maintains presence, 2.4-4x may still be suitable and acceptable.

9 Redirected Walking

A technique to create seemingly infinitely traversable virtual playspaces from roomscale environments. Redirected walking utilises imperceptible lateral shifts in the visual scene that cause the user to walk in an arc while believing they are walking straight ahead.

Facilitates longer virtual walking routes than straight trajectories, allowing users to walk through an effectively infinite virtual space. Users could be redirected by up to 18° over a 5m walking distance without being reliably aware, representing an arc with a 32m diameter.

Curvature and translational gain can be implemented concurrently without users noticing. Perceptual illusions, change blindness, or visual distractors can be used to reduce a users awareness of manipulators.

9.1 Problems

Requires large real world spaces, at least 5x5m, but often more to be effective. Reorientation events may be problematic for spatial understanding. Creating fully reachable VR spaces can be an issue, and there are largely unexplored implications for VR interactions.

9.2 Impossible Spaces

Leverage perceptual tricks like overlapping environments, redirection, vector, to create perception of a much more complex environment, e.g. bigger spaces or more complex interiors with many rooms.

9.3 Moving Platforms

Highly immersive, your movements are not altered, no interface to learn. Everything needs to involve these forms of movement to navigate environment. potentially less agency and less free exploration, and fixed paths.

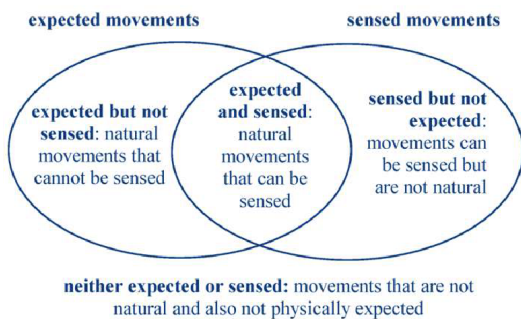
9.4 Non-Euclidean Spaces

Leverages visible or imperceptible portals. Again, highly immersive, your movements are not altered and there is no interface to learn. It makes it difficult to form a mental model of the world/environment/layout. Spaces tend to be dominated by cramped corridors with limited overview of the reachable/playable world by necessity - to ferry you from one space to the next. With a big enough play area, this is easy to forget.

Spatial Interaction

1 Input Sensing

Several sensing modalities supported by modern XR devices; headset movement and orientation, body position, peripheral sensing, hand tracking, eye tracking, facial expression tracking.



1.1 Headset Movement and Orientation

Necessary for perspective-dependent rendering, virtual objects maintain a plausible appearance as the head moves. Uses Inertial Measurement Units, or IMUs - a combination of accelerometer, gyroscope, and magnetometer. Measures acceleration (x,y,z), angular velocity (x,y,z) and orientation (pitch, yaw, roll)

1.2 Body Position

Necessary for understanding where users are, e.g. in relation to user interface elements, virtual objects, etc. Usually optically sensed, some devices can sense absolute position and orientation (outside-in trackers in a known space) and other can sense relative movements and orientation *changes* (inside-out trackers in an unfamiliar space)

1.2.1 SLAM

Short for Simultaneous Localisation and Mapping, maps to an unknown environment and tracks position within it. Useful for absolute vs relative positioning, how do you know where the user is?

1.3 Peripherals

Often used as a proxy for hand tracking, gives richer interaction metaphors. Typically fuses IMU and optical tracking, with position and orientation relative to headset. Usually also has hardware controls for extra sensing, e.g. analogue sticks or mechanical buttons.

1.4 Hand Tracking

Enables fine-grained interaction with objects. More degrees of freedom, e.g., user can pinch. Again, can either be optically sensed via headset or use controller sensing, relative to controller grip

1.4.1 Via Controller

For example, valve index controllers, uses capacitive sensors on the handle, and infers finger pose from grip proximity

1.4.2 Via Sensor Data

Uses fingertip and joint position, and/or curl, flexion, abduction, etc. Together, gives you position and pose.

1.5 Peripheral vs Hand Tracking

Why not hand track for everything? There are some potential problems

- Occlusion and visibility - cannot track what you cannot see, compelled to use sensor friendly postures.
- Noise and imprecision - controller tracking is generally more robust.
- Increased resource demands - tracking hands is costly, higher resolution and frame rate.
- Appropriateness for experience - think arm vs finger scale.

1.6 Other Gesture Techniques

Beyond just direct reach out and touch manipulation, we have

- Indirect manipulation - using hand tracking to interact with objects that are out of reach, e.g. pointing at things in the distance to target them or mapping hand motion to virtual object motion.
- Postural gestures - performing a hand posture, e.g. thumbs up as a shortcut to functionality or non-verbal communication
- Motion correlation - matching the movement of a virtual object, robust against false positive recognition

1.6.1 Pinching

Cursor control is an indirect manipulation, hands are in one place, cursor is in another. Cursor motion is mapped to hand motion. Pinching has become 'the' gesture, because its easy for optical sensors, and its easy for humans - it is **expected and sensed**.

1.6.2 Pushing vs Pinching

Early spatial computing often used push gestures. A familiar and realistic interaction metaphor, we generally push buttons by moving or flexing fingers towards them. Realistic metaphors are not always ideally suited for XR. There is a lack of haptics, users have nothing to push against, sensing is less reliable, as hands are facing away from sensors, and there is interface ambiguity - where pinching has a clearly defined end point, pushing doesn't.

1.7 Eye Tracking

Also known as gaze tracking, detects where the user is looking, not exactly head direction, but instead focus of attention (head direction does not always imply focus of visual attention). Conversely, eye gaze can reduce need for head movement. Potentially more ergonomic if it reduces need for extraneous head movement for selection

Gaze can be used on its own for input, e.g. dwell interactions, where a user pauses on an object for an extended period. Or, pursuit interaction, where user follows a moving target for extended period.

1.7.1 Multimodal Gaze Input

Gaze is often fused with other input modalities to improve robustness, e.g. gaze and hand because users often look at objects before touching, improves robustness of the hand tracking. Eye gaze implied intent to interact, reducing false positive recognition.

In gaze-and-commit systems, use additional input modalities to confirm intent, e.g. look at buttons to target them, then use other modalities to confirm selection. Can improve reliability, dwell might falsely assume intent.

1.8 Beyond input

Benefits of eye tracking extend beyond input, we can know where others' avatars are looking for example. Gaze can be an important social cue, giving benefit to immersion and presence, as well as a heightened social experience.

Can also allow us to use foveated rendering, rendering at high resolution where the user is looking.

1.9 Face Tracking

The ability to track subtle face movement, use defined *facial action units*, e.g. jaw drop, open mouth left, tighten left eyelid.

1.10 Sensing Beyond Input

Sensing is not just about input, it improves embodiment and communication, body tracking gives us an understanding of body language, hand tracking give us an understanding of social gestures, same applies for eye and face tracking.

Important for XR productivity and social experiences, human-human interactions rely on a myriad of social signals.

1.10.1 Multimodal Input

Input modalities are not mutually exclusive, they are not intended to be used independently of others. By combining input modalities, we can benefit from the relative strength of each modality, 'use the right tools for the job'.

2 Interaction Techniques

An interaction technique is a combination of hardware and software that can detect when a person performs a specific action and generates an appropriate response. When designing interaction systems, you establish relationships between input sensing, system state, and device output.

2.1 Defining an Interaction Technique

Start by thinking about the relationship with input data. The same input data can affect environment in different ways, with controller position and orientation we can detect collisions with virtual objects within arms reach, or point towards virtual objects in the distance. Other forms of sensing can be utilised in different ways - the difference is in the details

2.1.1 Collision-Based Interaction

Virtual content responds when the user collides with it, e.g. when controller collides with a sphere or hand collides with a button. Requires users to get within reach of the content, e.g. purposely positioned near the user, e.g. in an environment that users can move around in. Ideally suited for physical interaction metaphors, e.g hitting a ball with a bat or punching a target with your hands.

When designing *collision-based interaction*, we need to consider where is the locus of interaction? is it the controller position, the palm of the hand, or the furthest forward fingertip?

We need to decide when a collision occurs, if its when the locus is inside the object (does 'inside' mean directly inside or abstracted to a bounding box?) or when it is close enough (how close is close enough)?

Another thing is to decide when the input event completes, upon collision, or after an additional action?

2.1.2 Ray-Cast Interaction

Content responds when user points at it, like using a laser pointer for example. Can interact from a distance, this is ideal when user has limited range of motion, and gives more ergonomic viewing angles when content is further away.

When designing *raycast interactions*, we need to consider where the ray originates from? the controller, an extended finger, something else?

What path does the ray follow, a straight line or a curve? and is there a disambiguation method e.g. lock ray or depth ray?

Is there any input assistance, e.g. snapping to targets? and when is the input event complete? after an additional action, or a dwell period?

2.2 Direct Manipulation

Virtual context responds when users directly act upon it, manipulate the digital in similar ways to the physical. e.g. physical actions performed on or with an object, like picking up and repositioning an object, or pulling a bow string with one hand whilst aiming with the other.

Direct manipulation leverages familiarity, users know and recognise how things should work, which can make it easier to figure out how to achieve goals.

In XR, direct manipulation can contribute to immersion/presence. Actions in the virtual world are similar to actions in the physical world. Virtual object behave realistically with reference to the physical world.

Typically implies collision, i.e. getting your hands 'on' a virtual object rather than pointing at it, or interacting with/through virtual tools.

3 Input Technique Design

When designing an interaction technique, you establish relationships between input sensing, system state, and device output. Design and implementation affect usability, its important to understand the implication of design and implementation choices.

3.1 Modelling interaction state

We can use finite state machines to model interaction, these model the possible states of the system, and the events that cause transitions between states.

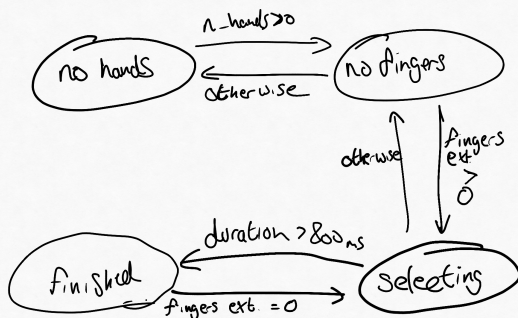
Modelling an interaction technique helps the design process, it formalises the behaviour that needs to be implemented, encourages reflection on how design choices affect usability, and identifies the events that might need appropriate feedback. Here is an example Now that we've identified system states and transition logic, have we covered all possibilities, or is there anything the user might do that the model doesn't cover?

Next, we should think about what happens in each state, how does the system change, what feedback should be given.

If you properly model the interaction, implementation is simple, states can be modelled as an enum, and transitions as a function.

4 Multi Sensory Experiences

Mixed reality is a multisensory experience. Visual sensation is predominant, in part because it helps sell the experience,



audio sensation is common, its cheap and easy, haptic sensation is still quite primitive, but theres a lot of current research into improved haptics for XR.

We integrate sensory information in search of a reliable explanation, based on available information, what is most likely happening right now. When sensory cues are inconsistent, our brain seeks reliability. When the body experiences something different to what it is seeing, it has to decide which of these competing signals is most likely.

4.1 Modality Dominance

Visual often overrides other sensory cues, up to a limit. Visual dominance is the tendency to favour vision as the most reliable.

Mixed reality diminishes awareness of physical reality. Creating opportunity to mask or manipulate sensory information. e.g. VR headsets occlude visual information, thus we think its an accurate representation of physical reality.

Inaccurate visual representations of reality can be useful, e.g. translational or rotation gain and redirected walking.

4.1.1 Haptics

Haptics refers to cutaneous sensations via the skin, e.g. roughness or temperature, or Kinaesthetic sensations via joints and muscles, e.g. proprioception.

When talking about haptics, people often mean vibration, but this is just one of the cutaneous touch sensations, it has no kinaesthetic component. Vibration is the predominant form of digital haptic experience.

4.1.2 Vibration

Vibrotactile output is the predominant form of haptics, creates a sensation of vibration, typically from a voice coil or linear resonant actuator. Most vibrotactile feedback is simple, on or off, but makes a positive contribution to the immersive experience - the sense that something is happening.

Vibration is a waveform, higher fidelity sensations *are* possible. We can improve platform support for sophisticated vibrotactile feedback, most toolkits now support waveform based vibration.

4.1.3 Wearable Haptics

Can deliver more targeted sensations, e.g. to individual fingers rather than full hand, including kinaesthetic force feedback from 'exoskeleton' type devices.

4.1.4 Haptic Retargeting

Use of passive haptics - arbitrary physical objects with similar features (size, shape, material qualities), essentially physical embodiments of virtual objects.

4.1.5 Haptic Proxies

Using generalised passive props to create touch sensations. In this instance, a multiplanar surface with several surface orientations. Users perceive just enough to enhance experience, and sometimes the feeling of physical content is enough.

4.2 Contrasting approaches

4.2.1 Vibrotactile

Perceptible and noticeable, hands presumably always holding device and so will feel it, but obviously requires peripheral interaction. Mimics contact with virtual objects, or tactile sensations of virtual objects - e.g. the 'click' of a button. There is still no resistance, you feel your hand hit something, but will move through it.

Has limited localisation, actuators will cause the whole device to vibrate, some controllers do have multiple though, e.g. meta touch pro.

4.2.2 Wearable

Also perceptible and noticeable, hands presumably always wearing device and so will feel it. Requires putting something on, which is fine in some contexts but less so in others. Similar application and haptic sensation as vibration, but some devices have the ability to render resistive force feedback

Better localisation, ability to render different sensations to distinct fingers.

4.2.3 Passive Haptics

Passive haptic proxies - come with inherent physicality, because there are actual physical objects. Potential mismatch between physical and virtual, creates convincing sense of physical contact, but may be less convincing for conveying material qualities. There needs to be some alignment between physical and virtual.

4.2.4 Ultrasound Haptics

Highly localised, typical focal points are 10mm in diameter. Can use several focal points and move them rapidly, or target different parts of the hand.

This is weak and often indistinct, conveys the feeling of touching something, sometimes conveys texture or material, but your hand moves through it.

Immersion, Presence: Subjective Experiences of XR/VR

The Goal of VR is to create the illusion of reality, we want to fool human perception by using technology to generate artificial sensation. In short, we want to **overcome the device**

gap - the point where we are aware that a device is mediating what we are experiencing and thus aware that the VR experience is not real.

But *why*? Because its engaging? fun? different? When is it a good idea, and for what purposes? Is it really needed, and what exactly about this illusion is needed?

1 Defining the subjective experience of VR: Presence

Presence refers to the subjective response to a virtual experience as if it were real.

*Presence is a **psychological** state [...] in which even though part or all of an individuals current experience is generated by [...] technology, part of all of the individuals **perception** fails to [...] acknowledge the role of the technology in the experience*

Presence has a long history but it is still an actively researched and highly debated concept with evolving definitions and scope.

1.1 Immersion

Immersion has often been used to refer to both objective capabilities of a system and to a subjective experience, e.g. *This system is really immersive / I felt immersed in this experience.*

The modern consensus is to define immersion as a technical, objective specification of an immersive system. More sensory stimulation/replacement = more immersive. A **fully** immersive system would replace and simulate all sensory channels.

1.2 Immersion vs Presence

Immersion is the extent to which technology enables to deliver a vivid illusion of reality to the senses of a human participant.

Presence is a state of consciousness, the psychological, subjective sense of being in a virtual environment/mediated space

Immersion produces a sensation of presence, or at least contributes to it. With VR, we aim to use high immersion to achieve high presence. Leading to users feeling like they *really are* in the virtual environment.

Immersion does not necessarily lead to presence, strictly speaking, more immersion $\neq$ more presence. It does usually contribute to it, individual traits highly contribute to presence, e.g. immersive tendencies, emotional attunements to certain situations.

Affordance View: An immersive system with specific capabilities **enables** specific types of illusions. A more immersive system, e.g. VR, is not inherently better than a less immersive one, it is a *qualitatively different experience*.

1.3 Benefits of High Sense of Presence

- Leads to greater engagement, excitement, and satisfaction - presumably leading to other effects (learning, retention, ...)
- Users more likely to behave like in the real world - people scared of heights irl will be scared in VR, necessary in some applications, like exposure therapy.

- More natural communication - use same cues as face to face conversation, envisioned as being the ultimate mediated communication tool.

When wanting to create a strong sense of presence, we want to use multiple elements that impact presence to create a rich multisensory experience. We want to have social actors/agents that interact with the user, and have the environment and experience respond to the user. But, what influences presence?

- Vividness - ability to provide rich experience
- Using a virtual body - users can see themselves
- Internal factors - some individuals simply have higher tendency for presence
- Interactivity - how much users can interact
- Sensory, realism, fidelity factors.
- among others...

Factors	Guideline
Ease of Interaction	Provide seamless interaction such that users can readily orient in, traverse in, and interact with the virtual environment.
User-initiated control	Provide immediacy of system response, correspondence of user-initiated actions, and a natural mode of control.
Pictorial Realism	Provide continuity, consistency, connectedness & meaningfulness in presented stimuli.
Length of Exposure	Provide sufficient exposure time to provide VE task proficiency, familiarity with the VE, and sensory adaptation.
Social Factors	Provide opportunities to interact with and communicate with others verbally or by gestures. Provide confirmation that others recognize one's existence in the VE.
Internal Factors	Identify the types of individuals who will use a VE system and their preferred representational system (i.e., visual, auditory, kinaesthetic).
System Factors	Providing stereopsis, head-tracking, a large field of view, increasing update rates, multi-modal interaction, and ergonomically sound sensors/ effectors facilitate presence.

1.4 Measuring Presence

Presence is a highly debated, subjective conscious experience. There is a lot of research on how to measure it. Some say that subjective measures of presence should be relevant, sensitive, convenient, nonintrusive, and reliable; others state that ideal measurements would be reliable, valid, multi-level sensitive, and objective; and some state the above and that it should also be measurable contemporaneously, continually, without modification to the scenario, and should be generalisable across VEs.

No measure of presence yet exists that meets all these criteria.

- **Subjective measures** - Self-report questionnaires, continuous measures, e.g. presence slider/drawing presence
- **Objective measures** - Behavioural (reflexes/flinches/startle responses), Physiological measures (heart rate, skin temp, skin conductance)

1.5 Breaking Down Presence

One of the first presence models divided presence into two main illusions/factors, **place illusion(PI)** - the feeling of

being there, aka spatial presence - and **plausibility illusion(PsI)** - the credibility of events in comparison with what would be expected in reality in similar circumstances.

Skarbez et al. tried to unify different presence models and terminologies, including PI/PsI as well as co-presence/social presence illusions. They present PI as being bottom-up(sensorimotor) and PsI as being top down(cognitive)

2 Key Illusions of VR/XR

Modern views on presence converge on a framework of 4 separate illusions/effects describing the subjective dimensions of immersive experiences:

Key Enabler	Illusion	Interpretation
Natural sensorimotor contingencies	Place Illusion (PI) <i>Spatial Presence</i>	<i>I am here</i>
Responsive, believable, congruent events	Plausibility Illusion	<i>This is really happening</i>
Bodily multi-sensory integration	Body Ownership Embodiment	<i>This is my body</i>
All of the above, with representations of other beings	Social Presence Co-presence	<i>I am here with others</i>

2.1 Concepts - Place Illusion (PI)

Place Illusion (PI/ π) refers to the illusion of being in a virtual place, even though you know for sure that you are not really there. It is a strong illusion, "bottom-up", it is the result of fast, automatic, sensory information processing. You cannot stop it by knowing it is false, you can only stop it by shutting out sensory input.

It is not difficult to achieve, if you perceive an environment through sensorimotor contingencies approximately the same as in physical reality, PI will occur, like many 2d visual illusions. It leads to feelings, qualia, of "being there, but knowing you are not there". Qualia are subjective, conscious experiences.

2.2 Sensorimotor Contingencies

How do we get a sense of being where we are, in real-life? Through the link between our perception and movement/action. If i move my head to the left, i expect certain changes in what i see and hear, if i move my arm to touch water, i expect to feel a certain haptic sensation.

these relations between motor actions and associated sensory sensations, learned throughout life, are called sensorimotor contingencies. These rules vary across senses. "Enaction" psychologists claim sensorimotor interactions are how we make sense of the world

The success of an immersive system in inducing PI will be determined in great part by the amount, and extend to which it supports the natural sensorimotor contingencies. Reproducing natural sensorimotor contingencies in VR (or failing to) is a huge determinant of the overall success of inducing place illusion/

2.2.1 Partial Ordering of Immersive Systems by SCs

Immersive systems can be *in principle* be put into partial orders. A first order system has SCs that simulate those of

physical reality.

A system of order $i < j$ can be used to simulate the SCs of j , but j cannot be used to simulate the SCs of i . Any immersive system of whatever order will have a set of SCs that enable perception of the virtual world within that system. A VR system might support natural SCs through head tracking (1st order), a desktop system might use joystick and keys(2nd order).

2.3 Breaks in Place Illusion

When people perceptually probe, they will inevitably break PI. Could be anything from latency failure in head tracking, looking at something closer than resolution permits, graphical glitches, touching something and feeling nothing, feeling something that should not be there (cables, weight of HMD).

SCs provide us a framework in which PI might occur, but it is not deterministic. In practice, PI recovers quickly from breaks.

2.3.1 PI in Non-Immersive Systems

If we consider PI to be bound to normal, real life SCs for perception (1st order system), then we can **never** have PI in desktop/tv based video game systems.

A 1st order system has SCs that simulate those of physical reality, you can pull up a chair, throw a ball. This happens in video games, but the type of SC is different, to look around, you use a mouse/controller instead of turning your head.

There can still be narrative immersion in games,books, or other types of non-immersive systems, but not actual PI.

2.4 Plausibility Illusions

Plausibility Illusion (PsI) corresponds to your subjective sense that observed events are real, i.e "this is really happening" It is the illusion that situations and events are what they appear to represent. They are really happening as they appear to be. This can apply in real life, theatre, cinema, etc, and in VR

Plausibility Illusion is hard, the slightest thing can break the illusion, it is also complex, it is multi-layered, combining both bottom-up and top-down processes.

2.4.1 Breaking Down PsI

In Skarbez model, Psi is seen as being top down (cognitive). In other words, it involves a slow, cognitive process requiring cognitive resources. Events happening in the simulation are compared to expectations - related to the suspension of disbelief.

Plausibility illusion has perhaps been the least researched so far. Plausibility emerges from the evaluation of information processed by sensation, perception, cognition, layers, and whether this information is congruent with expectations.

2.5 Embodiment

the sense of embodiment has 3 components, **Body ownership** - the virtual body is my body, **Agency** - i control the virtual body as my own, **Self-location**, ones self is located inside the body of this avatar.

Regarding appearance, the more anthropomorphic, the more ownership felt, its still possible to induce sense of body ownership towards avatar differing from real body.

With control, better visual motor tracking/synchrony gives more agency, in terms of PoV, first person gives a better sense of self location.

2.5.1 Proteus Effect

Yee & Baileson coined this effect of VR avatars affecting our behaviour as the proteus effect. VR users have the power to instantly/easily change the representation of their self, i.e. by changing their avatars. topic

The underlying assumption is that changing the body can change the mind, leading to overcoming bias, cognitive changes, pain reduction.

2.5.2 VR for embodying different perspectives

Exploration of VR for promoting perspective-taking in violent conflicts. Claims increased empathy 5 months post-experiment.

2.6 Co-Presence, Social Presence

One shortcoming in many common conceptualisations of presence is that they don't include aspects of social interaction. Yet VR is touted at a revolutionary communication medium that will radically change the way people interact with each other.

Co-presence and social presence are concepts to describe shared experiences in mediated environments. Like presence, there are many competing/contradictory definitions, and its still an actively debated area.

Co-presence - Essentially, feeling present with others in the VE. Defined as,

co-presence exist[s] when people sense that they are able to perceive others and that others are able to actively perceive them...rendering persons uniquely accessibly, available, and subject to one another.

Social presence - the degree to which a person is perceived as a 'real person' in a mediated communication.

The degree of salience of the other person in the interaction and the consequent salience of the interpersonal relationships.

2.6.1 Social Presence

What makes a person appear real?

- Interactivity
- Visual appearance
- Non-verbal cues
- Audio cues
- Touch
- Contextual cues

3 Agents and Avatars

Avatars are virtual representations of users, form/embodyment need not be humanoid. These are driven by a user, not autonomous.

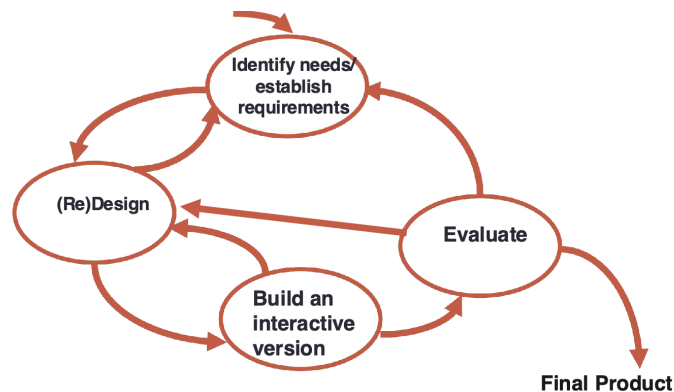
Agents, or Intelligent Virtual Agents (IVAs), or Embodied Conversational Characters (ECAs), virtual humans... Are autonomous, computer controlled. simulated social beings. They have many roles and contexts, like NPCs, guides, tutors, interface agents, etc. They are capable of expressing and recognising verbal/non-verbal behaviour.

Design and Evaluation for XR

1 Interaction Design

Designing interactive products to support people in their everyday working lives

Interaction design is the design of user experience with technology. It involves answering three questions, what do you do? What do you feel? and What do you know?



1.1 Needs Analysis

Create a deep understanding of the user and of the problem space. Understanding how XR can help address the user needs. Key questions to ask:

- Who is the user? Identify the different types of users
- What are the users need? Understand the users and look for insights
- Can XR address these needs? XR cannot solve all problems and is not the best choice for all problems

The different types of users are the *primary users* - regular users of the system, *secondary* - people providing support for the XR system, and *tertiary* - people providing funding/space for the XR system

We identify user needs by

- Learning from people - ask questions (questionnaires, interviews), run focus groups, observe people performing tasks
- Learning from experts - experts have in depth knowledge, and can give a lot of info in short time, look for existing process/problem documentation, and choose participants with domain expertise.
- Immersing yourself in context - put yourself in position of the user (role playing, day in the life, cultural probes), observe the problem space around you.
- Seek inspiration in analogous settings - take inspiration from different context than problem space, e.g. library redesign by going to apple store, similar scenarios in different places.

From understanding the user, look for needs, could be human emotional or physical necessities, these needs help define your design. Identify needs from user traits you noted, or from contradictions between information.

1.2 Design

Once user needs are found, solutions can be proposed, generate ideas through brainstorming, lateral thinking, formal problem solving, etc.

Elaborate on ideas and reduce to final design direction. Generate solutions, decide on the ones worth pursuing, repeat.

Design Considerations:

- Use UI best practices
 - General UI design principles can be applied to XR
 - Provide interface feedback - mix of reactive, instrumental, and operational feedback
 - Use constraints to specify relations between variables that must be satisfied.
 - Support two handed control
- Use of interface metaphors/affordances - design interface objects to be similar to physical objects the the user knows how to use, makes learning interface easier and more accessible
- Design for humans - use human information processing model
- Design for different user groups - different users may have different needs
- Design for the whole user - social, cultural, emotional, physical, cognitive

Typical VR Interface metaphors are direct manipulation (reach out and grab objects), ray casting (select object through ray), vehicle movement (move through VR environment through standard vehicle movement)

1.2.1 Affordances in XR

Just like anything else, in XR we need to design interface objects to show how they are used. Use visual cues to show possible affordance, perceived affordance should match actual affordances.

Design for perception	Design for Cognition	Design for Physical Ergonomics	Design for Different User Groups
Need to understand perception to design VR	Design for Working and Long term memory	Design for the human motion range	Design for different ages
Visual perception (many types of visual cue)	Working memory short term and limited, long term memory - recall by associative cues	Avoid gorilla arm syndrome from holding arm pose	Prior experience with XR systems
Auditory System (binaural cues, vestibular cues)	Model of current state of users environment (way-finding, spatial awareness, etc.)	Design for hand input	People with different physical characteristics
Somatosensory (haptic, tactile, kinesthetic, proprioceptive)	provide cognitive cues to help with situational awareness (landmarks, map knowledge, ect)	Consider bi-manual manipulation	Perceptual, cognitive, and motor ability.
Chemical sensing system (taste, smell)	support both ego and exo centric views	coarse and fine scale motion	

1.3 Prototyping

1. Find the quickest path to experience

2. Doing is the best kind of thinking
3. Use materials that move at the speed of thought to maximise your rate of learning

How can we quickly prototype XR experiences with little to no code?

First, *why* do we prototype? To create a quick visual design, capture key interactions, focus on user experience, communicate design ideas, and "learn by doing/experiencing"

From creating sketches in early ideation stages of design, to creating prototypes to capture and detail the actual design. The primary differences in a sketch vs a prototype is in the intent.

Typical development steps increase in fidelity and interactivity, starting from sketching, to storyboards, to UI mock-ups, to interaction flows, to video prototypes, to interactive prototypes, finally to native applications.

1.3.1 VR Prototyping Tools

Low fidelity - Sketched paper interfaces/storyboards, 360VR tools (InstaVR) a web based tool for simple interactivity, or In-VR prototyping, using MS Maquette/Sketchbox/Gravity Sketch

High Fidelity - Entiti: template based VR with visual programming, A-Frame: web based HTML VR tool, EditorVR: unity wrapper, Unity/Unreal game engine(s)

Sketching is not about drawing, it is about design, it is a tool to help you express, develop, and communicate design ideas fast.

1.4 Evaluation

Evaluation is concerned with gathering data about the usability of a design or product by a specific group of users for a particular activity within a specified environment or work context.

We evaluate *during* design and development, not after the product has been developed. This way, we find and rectify problems early, however it comes at higher evaluation cost and longer development time.

4 evaluation paradigms, *quick and dirty*, expert/predictive evaluation, usability testing, and field studies.

1.4.1 Quick and Dirty

A *quick and dirty* evaluation involves informal feedback from users or consultants to confirm that their ideas are in line with users needs and are liked. These are done at any time, and emphasis is placed on fast input to the design process rather than carefully documented findings.

1.4.2 Predictive Evaluation

Experts apply their knowledge of typical users, often guided by heuristics, to predict usability problems. Can involve theoretically based models. A key feature of predictive evaluation is that users need not be present. This is relatively quick and inexpensive.

1.4.3 Usability Testing

Recording typical users' performance on typical tasks in controlled settings. As the users perform tasks they are watched, recorded, and their inputs are logged. User data is used to

calculate performance times, errors, and help determine system usability. User satisfaction questionnaires and interviews are used to elicit users opinions.

1.4.4 Laboratory-Based Studies

These can be used for evaluating the design or system, are carried out in an interruption free usability lab, can accurately record some work situations.

Some studies are only possible in a lab environment, some tasks can be adequately performed in a lab. Lab-based studies are useful for comparing different designs in a controlled context.

1.4.5 Field/Ethnographic Studies

Field studies are done in natural settings. The aim is to understand what users do naturally and how technology impacts them. In product design, field studies can be used to identify opportunities for new technologies, determine design reqs, decide how to introduce new technology, evaluate technology in use.

1.4.6 Characteristics of Approaches

	Usability testing	Field studies	Predictive
Users	do task	natural	n/a
Location	controlled	natural	anywhere
When	prototype	early	prototype
Data	Quantitative (+mixed)	qualitative	problems
Feed back	measures & errors	descriptions	problems
Type	applied	naturalistic	expert

2 XR Evaluation Studies

Typical research questions might include:

- Is collaboration with AR/VR/XR better than video conferencing?
- What is the impact of a particular XR input method?
- How should people be represented in Social XR interfaces?
- What communication cues should be added to improve collaboration?
- How can you effectively support collaboration in hybrid/immersive interfaces?

2.1 Pilot Studies

A small trial run of the main study, this can identify issues with the interface design. A pilot study checks that the evaluation plan is viable, that you can conduct the procedure, and interview scripts, questionnaires, experiments, etc. work appropriately. Pilots are used to iron out problems before the main study

2.2 Subjects

The choice of subjects is critical to the validity of the results of an experiment. Subjects group should represent expected user population. Factors such as age group, education, skills,

culture, tech background should all be considered. The sample size should be large enough to be statistically representative of the user population

2.3

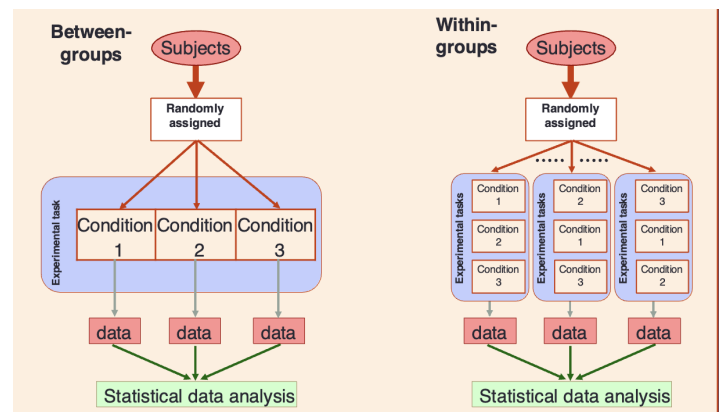
Hypothesis and Variables A hypothesis is a prediction of the experiment outcome. Experiments manipulate and measure variables under controlled conditions.

There are two types of variable, independent, meaning variables that are manipulated to create different experimental conditions, and dependent, meaning variables that are measured to find out the effects of changing the independent variables.

2.4 Experimental Methods

It is important to select the right experimental method so that the result of the experiment can be generalised. There are two main experimental methods, between- and within-groups

In a between groups study, each subject is assigned to one experimental condition. In a within groups, each subject performs under all the different conditions.



2.5 Data Collection and Analysis

The choice of a method is dependent on the type of data that needs to be collected. In order to test a hypothesis, the data has to be analysed using a statistical method. The choice of statistical method depends on the type of data collected. All the decisions about an experiment should be made **before** the experiment is carried out.

2.5.1 Data Types

- **Qualitative|Subjective** : Subjective survey, i.e. Likert scale, condition rankings. Observations, and interview responses
- **Quantitative|Objective** : Performance measures, i.e. time, accuracy, errors. Behavioural/physiological (EDA,ECG,EEG)

2.6 XR Questionnaires

The main questionnaires relevant for most XR experience measurements:

- Cybersickness: SSQ - simulator sickness questionnaire
- Workload: NASA-TLX - NASA task load index
- Fatigue: Borg CR-10 - perceived exertion
- Presence: IPQ - Igrou Presence Questionnaire
– Or Slater-Usuh-Steed (SUS), or Presence Questionnaire (PQ)

- Social Presence: Biocca's Networked Minds Social Presence Inventory
- Embodiment: General Embodiment Questionnaire

2.6.1 SSQ

Simulator Sickness Questionnaire - reference for cybersickness. Derived from general motion sickness questionnaires.

2.6.2 NASA-TLX

Perceived workload on several sub-dimensions, e.g. mental, physical, etc.

2.6.3 Borg CR-10

Fatigue/Exertion scale. Used to evaluate 'exergames', XR weight perception

2.6.4 IPQ

General Presence Questionnaire. For spatial presence (the sense of being physically present in the VE), Involvement (measuring the attention devoted to the VE), and experienced realism (measuring the subjective experience of realism in the VE)

2.7 Beyond Questionnaires

We can capture data during experiment instead of after, moving from performance measures to behavioural, process measures. We can capture richer types of data, like physiological cues (EEG, GSR, EMG, Heart rate) and behavioural cues (body motion/positioning).

Gain a higher level of understanding, mapping data to emotion recognition, cognitive load, etc. and use better analysis tools - Video analysis, conversation analysis, multi-modal analysis.

For example, we can use sensors in HMDs to achieve this, for eye tracking, heart rate, etc.

Bystander & Reality Awareness

Virtual reality introduces significant barriers to a users awareness of and interaction with nearby people and the surrounding environment. This lack of awareness results in problematic scenarios between the VR user and their surrounding environment.

1 Why do we Need Reality Awareness?

Potential for problematic interaction with nearby (co-located) bystanders, or with nearby objects and furniture, or pets.

Interacting with reality while in VR is generally problematic, this is why we should stay in the designated "safe" space to use VR in (the play space). Even so, In VR, it is problematic to use a keyboard, pick up and eat/drink food or drinks. More abstractly, we don't know how much time has passed, you cant see changes in natural daylight.

We want a VR display capable of identifying and understanding elements of the users surrounding reality and responding to it. You make the VR display aware of and react to reality.

The Quest 3, for example, is a device capable of presenting both VR and AR content, it is a device capable of varying the amount of augmentation presented to the user. We can provide the VR user with some information about reality and/or their environment in real-time to increase their awareness.

1.1 Boundary Awareness

A classic example of reality awareness is **boundary awareness systems**. In short, VR users sometimes collide with walls, the solution is boundary awareness. Get the VR user to outline a safe area, and inform the user when they are too close to the edge - but how?

Most systems use virtual walls, or enter a pass-through view when getting too close/past a boundary. Pass-through is more informative but more disruptive to the experience. Could also use haptic/audio notifications, but these might also be problematic.

2 Bystander Awareness

Boundaries are boring. Boundaries are easy to solve. People are interesting. People are complicated. Boundary awareness is checkers. Bystander awareness is chess.

- Joseph O'Hagan

People are unpredictable, and while the VR user is fully augmented in VR, the bystander is interacting with the real world. How might an interaction go? Are interactions effective? How do they breakdown? How can reality awareness enable a better interaction?

2.1 The Role of the Bystander Awareness System

Is it to allow the bystander to easily interrupt the VR user? No, not really. Is it to enable an improved interaction between the bystander and VR? Yes, but how?

We need to first understand how bystander interactions occur with VR users

2.2 Interrupting a VR User

Understanding "how interactions occur" is a big problem, there are infinite possible interactions. We know that interruptions are a key part of them, the bystander must interrupt a VR user who is unaware of them in order to interact with them.

By understanding how interruptions occur, we understand comfort enacting an interruption, and if systems should be developed to ease this process, we understand how interruptions occur. Through understanding this, we can start to understand what a bystander awareness system should do.

Most bystanders are comfortable interrupting the VR user using speech and/or touch, in some cases both - speech to announce their presence, touch to inform where they are in the environment. This is likely to surprise a VR user, mostly because of unexpected touch.

Sometimes the VR user has headphones on, so they miss the speech, and sometimes physical collisions can occur when going for touch.

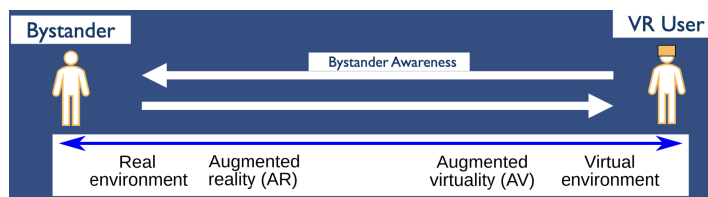
If you give alternatives, most people *don't care*, speech and touch are fast, existing norms. Other options might be interesting and nice to have, not not essential.

2.2.1 Waiting to Interrupt

Bystanders might wait for an optimal moment to interrupt, waiting until they hear/see the VR user is in a moment of downtime to then interrupt.

2.3 The Role of Bystander Awareness in Interruptions

Primarily, its not so that its easier for the bystander to interrupt, its bringing the VR user to a level playing field. Without bystander awareness, the bystander has all the power and can exploit the VR user, with it, the VR user can interrupt the bystander - acting as a safety feature.



3 Interacting with a VR User

Again, there are infinite possible interactions which could occur.

Speaking is problematic, the VR user doesnt know the person is there. Lets say the VR just cant properly hear them and takes off the headset.

Two points emerge as clear problems to address:

- Establishing bystander existence (and position) in the surrounding environment.
- Facilitating verbal interactions between bystanders and VR users

4 Designing Bystander Aware Systems

Lets assume the bystander walks into a room and the VR device detects/captures their existence. How would you inform the VR user of a bystanders existence?

The 3 main approaches are through text notifications, avatars, and a pass-through view.

The problem with a generic text notification is that if you don't tell the user **who** is in the room, or **where** they are, some will take off the headset to check, this isn't what we want from awareness systems. Clearly, the identity and position of the bystander is often important.

We could use a generic avatar, but there's no identifiability, pass-through would be better. If we can embed information about identity and text into text notifications, this could also be good.

4.1 Non-Visual Solutions

Those were all visual solutions, but what if we wanted to use audio or haptic approaches. Could be problematic, there could be mistaken for part of the VR experience. Compounded with the inability and ineffectiveness at conveying

the identity and position of the bystander, these are not as good as visual.

However, we can use aural awareness to a different end. We can facilitate aural awareness to enable conversation - suppose you can detect when a bystander *talks* to the VR user, how would you increase the VR users awareness of what they're saying?

2 main approaches for this, dynamically lowering the in-VR audio volume, or dynamically removing audio elements.

4.1.1 Dynamic volume lowering

Simple, you dynamically lower the volume of the audio when you detect a conversation so the VR user can better hear the bystander. Its simple, it works, and people like it.

4.1.2 Dynamic removing audio elements

You remove increasing amounts of audio components from the in-VR audio. The idea is that VR is a visual heavy medium in terms of creating a sense of presence, so we remove audio to facilitate verbal interaction.

This works, in theory, but its harder to achieve. Dynamic volume adjustment is a universal solution across applications, audio removal requires tuning on a per application basis.

Being Social in XR

Over the past few decades, we've went from humans commanding very specialised machines, to people having personal home computers, to most people having mobile computers (smartphones) and (maybe soon) to immersive computing, where computers change how people experience the world.

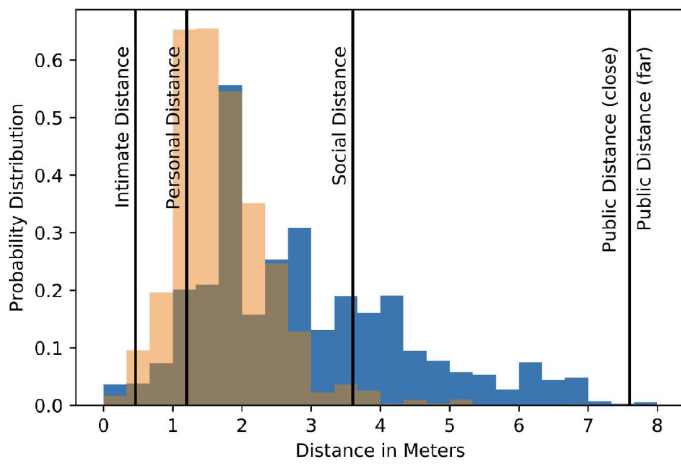
In such a world, virtual content would be indistinguishable in every way from physical content. As such, we need to make interpersonal interaction in XR work, otherwise these technologies could fragment and divide our experiences.

1 VR Outdoors

In 2017/18, there was a glasgow short film festival with a pop-up 360 cinema, where people could experience movies in VR in a social setting. Some people are in VR, other arent. Socially, this situation works because its a time and place where it is expected and accepted.

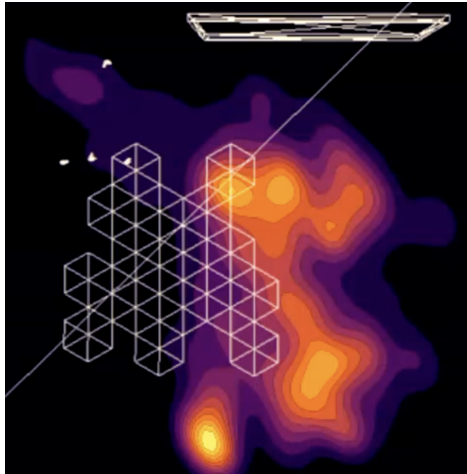
2 Social VR

We can track where people move/stand in VR environments to see how they act in comparison to how we would expect them to act in the real world.



An analysis of a virtual keynote session, which also had a breakout room session, shows that in the keynote, the larger group activity, people stood further apart, as expected, and in the breakout room, tended to group closer together.

While its completely possible in VR to stand in the same place as another, to overlap the space you take, people rarely stood within 1m of each other, as you might expect in real life. Just like in real life, if you stand too close to someone, they will likely move away.



Again, looking at this heatmap of where people stood, no one place was too dense, but people did tend to stand together. Interestingly, not many people stood over the floor decal, preferring to stand beside it.

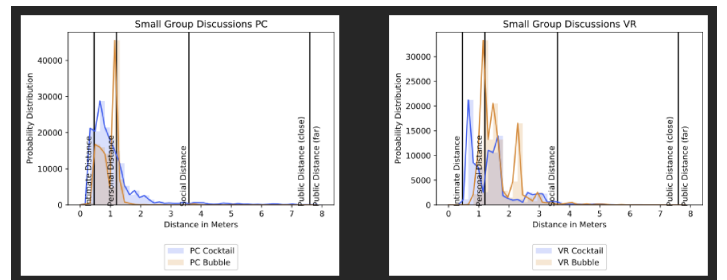
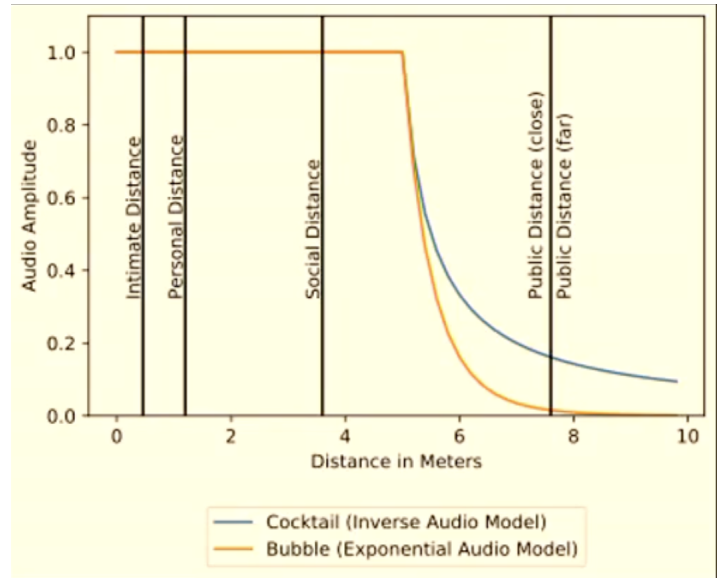
Proxemic interactions are congruent with those in physical environments, even though VEs are not constrained by physics. Instrumented approaches can provide insights into the effect of environment texture, shapes, and design. Quantitative data can answer questions on social behaviour, such as group formation, social translucence, and social cues.

2.1 Digital Proxemics

Digital Proxemics describe how we use space in virtual environments, and how the presence of others influences our behaviours, interactions, and movements.

How much of an impact does VR have over desktop display? and comparing inverse versus exponential audio

In the PC example, there are more collisions in the intimate zone - there is less sense of discomfort being too close to people on a 2d display. In VR, intimate zone has very limited collision, personal zone has fewer too, and most is in social distance



In desktop, people don't often turn to look at the active speaker - a behaviour very unlike real life. In VR, this does happen. People more closely mimic behaviours of showing that they are paying attention to the speaker.

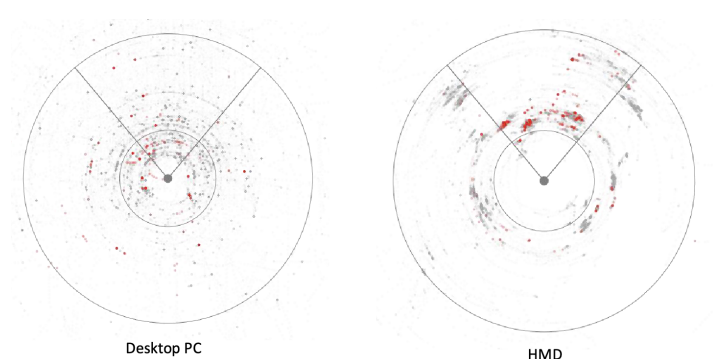
2.2 Key Takeaways

Designing effective experiences for virtual environments requires an intricate knowledge of how people make use of and behave in virtual spaces.

Although this research only scratches the surface of the broader challenges, we hope this inspires future research into digital proxemics and improves how we collaborate and socialise in virtual environments in the future.

Social and Ethical Challenges of XR

Usually broken down into Social Acceptability, Privacy, Censorship, and Perceptual Manipulation.



1 Social Acceptability

Social acceptability is how a user interface is perceived by the user and by other people. A concern for mobile interaction because of varying usage contexts - different places, people, social norms, social settings, etc.

Multiple perspectives to consider, how comfortable does the user feel? for a spectator, does interacting appear normal?

1.1 Interaction Visibility

Compare input vs output visibility. Input actions can be highly visible, e.g. gestures, speech, and output effects might be (near) invisible, e.g. a small screen or haptics.

Input might draw unwanted attention, e.g. with unusual actions like tapping on face, waving hands around; or the actions could be disruptive, like speech input, clicking fingers, or tapping on surfaces.

1.1.1 Situation/Context

It all depends on the situation in which an interaction occurs, is there an audience? is it the general public, colleagues, strangers, family & friends? Is the interaction taking place in public, at home at the office, on a train, on a street?

Acceptability depends on the combination of location and audience an interaction occurs. something done at home alone might be a lot less acceptable in public around strangers.

1.1.2 Interaction Design

It also depends on interaction design. Gestures where the hands are near your torso, with small movements, and have a short duration are more acceptable than the alternatives.

1.1.3 Device Location

Also a factor, where is the device (and what is the device design)? Devices on the wrist, forearm, or in the hands are more acceptable than something on the neck for example.

1.1.4 Familiarity

The spectators familiarity with the devices can also play a part, smartphones are more established, and have more easily explainable user behaviour. Smart glasses are less familiar. Attitudes can and do however change.

1.2 Social Acceptability

Impacted by interaction, visibility, spectator familiarity, device visibility, location/context, perceived device capabilities. This perception of socially acceptable or unacceptable devices is also partially related to privacy and perceptual harm concerns.

2 Privacy

Privacy is the ability of an individual or group to seclude themselves or information about themselves, thereby expressing themselves selectively.

Privacy is the claim of individuals, groups, or institutions to determine for themselves when, how, and to what extent information about them is communicated to others. Each individual is continually engages in a personal adjustment process in which

they balance the desire for privacy with the desire for disclosure and communication

Westinn defined 4 states of privacy, as **Solitude** - the state of being free from observation by others, **Intimacy** - the seclusion required for a close associates, **Anonymity** - The condition of being unknown and free from identification, and **Reserve** - the desire to limit disclosure to others

2.1 Why Privacy is a Concern for Mobile Computing

Mobile devices travel with us, they are worn, active, and used throughout the day. They contain a breadth of sensing capturing data about our context, physiology, attention, biometrics, and more. In many cases, they can capture information about others nearby.

This data can be processes and used to discern and capture information about individuals that they may not consent to, or be aware of. Can be used to understand habits, likes, dislikes; reveal personal information about health characteristics, sexual orientation; break anonymity, tracking individuals and their associations.

- **Solitude** - Your own and other devices can observe you through cameras, microphones, etc.
- **Intimacy** - Your mobile devices are always with you
- **Anonymity** - A wealth of biometric data about you is being captured
- **Reserve** - From device/app permissions to legal constraints

2.2 Privacy Risks of XR

A seemingly consensual erosion of privacy from the XR user, and a non-consensual erosion of privacy from the bystanders to the device.

Users lose physical and mental privacy as a result of required XR sensing (cameras, microphones, IMUs, gaze, etc.) Access to this data would allow for immediate comprehension of daily life and longitudinal understanding of behaviours.

2.2.1 Risk to Bystanders

XR Devices can selectively enhance our sense through augmented perception, sometimes good for overcoming impairments, but also can be abused for eavesdropping or recording, going beyond the natural limits of human memory and sensory reach.

One or more XR devices in a space have the capacity to surveil the space completely, supporting persistent, ubiquitous tracking, recording, volumetric capture, etc.

2.2.2 Risk to Users

Perception of Reality can be altered, in subtle ways like behavioural nudging, altered/diminished reality, etc. or in not so subtle ways, like advertising, altering perception of sub-groups, reinforcing bias.

There is also a risk to identity and anonymity, behavioural biometrics and be used to identify someone to a reasonable degree of accuracy.

2.3 Rejection of Technology

Privacy concerns can lead to rejection of technology. Google glass for example was initially condemned over concerns was always recoding.

2.3.1 Protecting Privacy

First, is the legal protection. Currently largely reliant of existing privacy laws and voluntary restraint from market leaders. GDPR governs how personal data can be captured, stored, and processed. Protects personal data, but remains untested in XR scenarios.

There are also social norms, XR devices are not common enough or widely adopted.

3 Perceptual Harms

Everyday AR will empower users, communities, and governments to alter, augment, diminish, or otherwise intervene with our perception of reality.

Everyday AR doesn't need sophisticated AI to enact dangerous alterations of our perception of reality, basic heuristics is enough. Imagine how divisive politicians might use this, no longer influencing people's thoughts but their perception of the world and those around them.

3.1 Social Expression of Identity

Present a public-facing digitally augmented facade that adapts to who you are with, where you are, current context. Obvious benefits in supporting representation, but risks in application and agency.

3.2 Augmenting Shared Public Spaces

Space mediates community, creativity, and identity. Communities need the tools to co-create their own streets and spaces. When they are involved in creating and nurturing their own environments, they feel a greater sense of agency, community, and pride - **Collective efficacy**

3.3 Controlling Perception

Large companies might want to control what we perceive, but why? To influence...

- Attitudes - how you think or feel about something
- Decision making - the action or process of making a decision
- Memory - how we store and remember information
- Behaviour - how we respond to a given stimulus
- Affective State - your feelings and emotions
- Cognition - how you acquire and understand knowledge

Advertising is the obvious example, a route to subsidizing our access to and use of XR tech. A slightly more interesting area is *activism*

3.4 Decision Making

3.4.1 Deceptive Designs

Deceptive designs or dark patterns aim to influence a user's interactions with an interface to benefit someone other than the user. XR offers affordances well suited to such designs, e.g. the ability to render 3D content anywhere anytime, the ability to direct what users perceive, and the ability to understand the context and state of the user to personalise manipulations

3.4.2 VR Training

VR is increasingly being used by police forces for training, and that training often involves the use of firearms in high pressure scenarios. Is this training valid, do we trust that

this is effective and safe? What is the impact of this training? How is the training designed?

4 Rights and Controls

- **Right to metaversal access?** We could consider there being common metaverse(s) or metaversal layers that everyone has the right to access, and this might extend to fundamental capabilities around augmented intelligence and perception, minimising inequality in the augmented society.
- **Perceptual agency?** The right to augment the AR users sense-scape, what elements in reality are permissible to alter, augment, extend, or diminish.
- **Perceptual Autonomy?** The right to control what you perceive, i.e. who can mediate your perception of reality
- **Cognitive autonomy?** Tensioning the right to free-will and independence of thoughts, attitudes, behaviours, and actions against non-consensual manipulation through perceptual mediation

Ultimately, should we try to maintain a common objective reality we all experience? or should individuals, groups, governments be able to overwrite reality?

Privacy and Security in XR

1 Human-Centered Security & Privacy

Aim of human-centered security and privacy is to make privacy and security an integrated, natural, unburdened part of human-computer interaction. Key challenges are that security concepts are complicated for the average user, and security is a secondary task. Also, human capabilities are limited - people can't always remember multiple unique strong passwords.

People designing systems want to add more security to keep threats away, users just want products to be easier for themselves to use, this in turn makes it easier for attackers, e.g. using an easy to remember pin.

There are more and more threats everyday, from social engineering/phishing/side-channel/shoulder-surfing etc.

2 Computer Security

Protecting systems against misuse and interference to preserve **confidentiality, integrity, and availability** of system resources.

2.1 The CIA Triad

2.1.1 Confidentiality

Covers data confidentiality and privacy. Data confidentiality means that private or confidential information should not be made available or disclosed to unauthorised individuals. Regarding privacy, an individual's data should only be used/disclosed according to a set of rules.

Other related concepts are secrecy - to keep data hidden, and anonymity - to keep identity hidden

2.1.2 Integrity

Data integrity: Information and programs can only be changed via an authorised operation.

System integrity: systems consistently perform their intended function, without unauthorised manipulation intended or otherwise.

2.1.3 Availability

Systems and data are promptly usable to authorised users.

2.2 What About XR Safety?

In two parts, physical and psychological safety.

Physical safety from preventing collisions or falling, and defining safe physical boundaries (reality awareness), and reducing strain and discomfort from prolonged XR use, creating suitable device fit and interaction designs.

Psychological safety from preventing and addressing verbal and visual harassment in virtual spaces, and content moderation, i.e. filtering harmful or distressing content to protect users.

2.3 Threat Modelling

Perfect security is not possible. Attempting to protect against everything is too expensive, inconvenient, and not worth the effort – we need priorities.

Identify the most likely ways the systems will be attacked. Identify the likely attackers and their resources, and the consequences of the attack. Attackers will exploit the weakest link in the system.

2.4 Implications of XR

Users are not aware of what data is collected, nor the consequences of the collection, nor when the data is being collection. Users might not realise the value of their XR data.

We should facilitate awareness and transparency. Give users an understanding of why their data is needed. XR devices should be transparent about data capture and usage.

3 Privacy in AR

A problem with today's AR headsets is they are increasing in quality, quantity, and scope of data collection. They are equipped with always-on sensors and see prolonged use.

An alternative would be to let users decide how much data they want to provide to an AR application by showing them the impact to the user experience. For example, an application with the following levels of collection;

1. no microphone access
2. enough microphone access only to know that there was sound
3. aware of the occurrence and direction of sound
4. ability to transcribe sound (and give direction as before)
5. identify the speaker, transcribe, and give direction

AR privacy is not like privacy on other devices, there is a need for fine-grained permission, not a simple give or deny all permission.

Users want control, so we should support users in making the best decisions, being transparent lets users trust the app and device more.

A more informed user is good for everyone. But there is more burden on developers to write effective prompts/slider steps.

3.1 Physical Safety in VR

The problem with perceptual manipulators in VR, is that they are designed to be imperceptible - the users often do not understand they are being manipulated. This can be open to abuse.

The user could be tricked into installing malicious apps, which gets access to the data of the user and their environment. The attacker deploys manipulation techniques to provoke harm.

Classification of attacks:

- Puppetry attacks - walking, arm movements
- Mismatching - false positive/negative, swapping

Mitigations :

- Awareness and consent of users
- Validating apps
- Platform level mitigations
- Community guidelines

3.2 Psychological Safety in VR

Due to co-presence, immersion, embodiment, anonymity, there are new forms of harassment.

Despite age restrictions, 47% of children under 14 use social VR regularly (at least every two weeks). Parenthood familiarity with social vr, and presence of supervision tends to affect views of age appropriateness

Abuse and harassment does not only come from other XR users, bystanders in the same physical space might take advantage of the VR users lack of awareness of the real world in different ways.

Reality-Virtuality Continuum

Virtual Reality (VR)

"Virtual reality entails presenting our senses with a computer generated virtual environment that we can explore in some fashion." Completely supplants your visual and mostly supplants your auditory perception of reality.

Augmented Reality (AR)

Reality with some virtual elements. Adds digital information or objects to the user's view of the real world.

Extended Reality (XR)

An umbrella term for all immersive technologies including VR, AR, and mixed reality. The virtual supplements the real, i.e., the environment contains more information.

Mixed Reality (MR)

A subset of XR where real and virtual elements interact with each other.

Diminished Reality

The virtual removes the real (e.g., filters out unwanted objects from view).

Altered Reality

The virtual transforms the real (e.g., changing the appearance of real objects).

Hybrid Reality

The virtual completes the real (e.g., physical illusion boxes).

Extended Perception

The virtual translates imperceptible aspects of the real world (such as radiation or ultrasound) into information we can perceive with our available senses.

Spatial Computing

Computing paradigm where digital content exists in and interacts with the real world, often used to describe AR experiences that understand and integrate with physical space.

VR Hardware Types

Head-Mounted Display (HMD)

A display device worn on the head with small displays in front of each eye to deliver stereoscopic 3D visuals.

Inside-Out Tracking

Tracking system where sensors are on the headset itself, scanning the environment to determine position (e.g., Meta Quest).

Outside-In Tracking

Tracking system where external sensors in the environment track the position of the headset and controllers (e.g., Valve Lighthouse system).

6DoF (Six Degrees of Freedom)

Tracking that allows for movement in three dimensions (up/down, left/right, forward/backward) plus three rotational movements (pitch, yaw, roll). Full spatial freedom.

3DoF (Three Degrees of Freedom)

Tracking that only allows for rotational movement (pitch, yaw, roll) but not positional tracking.

CAVE (Cave Automatic Virtual Environment)

Room-scale projection-based VR environment where images are projected onto multiple walls of a room-sized cube.

Foveated Rendering

A technique that renders at high resolution only where the user is looking (using eye tracking) and at lower resolution in peripheral vision to save computational resources.

AR Hardware Types

Spatial AR/Projection Mapping

Uses projectors to display digital content directly onto physical objects or surfaces.

Heads-Up Display (HUD)

Transparent display that presents data without requiring users to look away from their usual viewpoint.

Passthrough AR

Uses cameras on a VR headset to capture the real world and display it inside the headset with augmentations (e.g., Meta Quest 3, Apple Vision Pro).

Optical See-Through AR

AR that uses transparent displays where the user can see the real world directly through the display (e.g., Microsoft HoloLens).

Retinal Display

Projects light directly onto the retina of the user's eye.

VR/AR Key Characteristics

Immersion

Being surrounded by a vivid illusion of a virtual world, conveyed across one or more senses, preferably in 3D and rendered correct to perspective/movements.

Presence

"A psychological state in which even though part or all of an individual's current experience is generated by technology, part or all of the individual's perception fails to acknowledge the role of the technology in the experience." The feeling of "being there" in a virtual environment.

Place Illusion (PI)

The illusion of being in a virtual place, even though you know for sure that you are not really there (also called spatial presence).

Plausibility Illusion (Psi)

The illusion that what's happening in the virtual environment is really happening, despite knowing it's virtual.

Interactivity

That your actions impact the virtual world realistically, and that you can interact with/react to the virtual world.

Field of View (FOV)

The extent of the observable world that is seen at any given moment, usually measured in degrees.

Latency

The delay between user input and system response, particularly motion-to-photon latency (time between head movement and display update).

Motion Parallax

The apparent difference in position or direction of an object viewed from different positions.

Interpupillary Distance (IPD)

The distance between the centers of the pupils of the eyes, important for proper stereo display calibration.

XR Sensing Technologies

SLAM (Simultaneous Localization and Mapping)

A technique where a device maps an unknown environment while simultaneously tracking its position within it.

Visual Position Systems (VPS)

Uses already acquired mapping data to identify shared landmarks and locate a device in relation to those landmarks.

IMU (Inertial Measurement Unit)

Combination of accelerometers, gyroscopes, and magnetometers that measure orientation, acceleration, and magnetic fields.

Time of Flight (ToF)

Sensing technique that measures the time it takes for emitted light to return after hitting objects, used for depth sensing.

Structured Light

Projects known patterns of light and analyzes their deformation to calculate depth and surface information.

Binocular Stereo Vision

Uses two cameras to calculate depth by analyzing differences between the images, similar to human depth perception.

LiDAR (Light Detection and Ranging)

Remote sensing technology that uses laser light to measure distances and generate precise 3D information.

Physiological Sensing in XR

- EOG (Electrooculography) - Non-camera based eye movement tracking
- EEG (Electroencephalogram) - Brain activity monitoring
- EMG (Electromyography) - Muscle activity sensing
- EDA (Electrodermal Activity) - Skin conductance measurement
- PPG (Photoplethysmograph) - Heart rate and oxygen detection
- FNIRS (Functional near-infrared spectroscopy) - Brain activity using IR light

Spatial Concepts

Egocentric Perspective

Describes an object's location with respect to the perceiver's perspective. Content moves with the user.

Exocentric Perspective

Describes an object's location independently of the perceiver's location or perspective. Content is fixed in world space.

Spatial Anchors

Reference points in the physical world that AR content can be attached to, allowing virtual content to remain in place.

World-Locked Content

Digital content that stays fixed in a physical location regardless of user movement.

Digital Proxemics

How users use space in virtual environments, and how the presence of others influences behaviors, interactions, and movements.

Locomotion & Movement

Vection

The illusion of self-motion induced by visual stimuli while the body remains stationary.

Cybersickness

Motion sickness in VR caused by sensory mismatch between visual and vestibular systems, where the visual system perceives movement that the vestibular system does not.

Redirected Walking

Technique to create seemingly infinitely traversable virtual spaces from room-scale environments by imperceptibly manipulating the user's path.

Teleportation

VR movement technique where the user instantly changes position by selecting a destination.

Perceptual Manipulators

Techniques that alter the user's perception, such as redirected walking, rotational gain, or translational gain.

Translational Gain

Amplifying real-world movements in VR, such as making a 1-meter walk in reality translate to 2 meters in VR.

Rotational Gain

Amplifying or reducing real-world rotational movements in VR.

Interaction Concepts

Raycasting

Interaction technique where a line/ray extends from a controller or hand to select distant objects.

Direct Manipulation

Interaction technique where users directly act upon virtual objects, manipulating them as they would physical objects.

Haptic Feedback

Tactile sensations provided to users through vibration, force, or other means.

Passive Haptics

Using physical props that correspond to virtual objects to provide realistic touch sensations.

Collision/Colliders

System that detects when virtual objects intersect, used for interaction and physics simulation.

Foveated Displays

Display technology that renders at different resolutions across the field of view, highest at the center where the eye is focused.

Social & Ethical Concepts

Social Presence

The degree to which a person is perceived as a "real person" in mediated communication.

Co-presence

The feeling of being present with others in a virtual environment.

Embodiment

The sense that a virtual body is your body, combining body ownership, agency, and self-location.

Proteus Effect

The phenomenon where the behavior of an individual within virtual worlds is changed by the characteristics of their avatar.

Bystander Awareness

Systems that make VR users aware of people in their physical environment.

Reality Awareness

Systems that provide VR users with information about their physical surroundings.

Social Acceptability

How a user interface is perceived by both the user and observers in social contexts.

Perceptual Agency

The right to augment one's sense-scape, controlling what elements in reality are altered.

Perceptual Autonomy

The right to control what you perceive and who can mediate your perception of reality.

Augmented Cognition

Supporting and enhancing cognitive state/user's mental model, decision making, and attention.

Evaluation Concepts

SSQ (Simulator Sickness Questionnaire)

Standard questionnaire for measuring cybersickness.

NASA-TLX (Task Load Index)

Measurement tool for perceived workload in interfaces.

IPQ (Igroup Presence Questionnaire)

Questionnaire measuring presence in virtual environments.

Sensorimotor Contingencies

The rules governing how sensory input changes with movement and action, crucial for inducing Place Illusion.

Vergence-Accommodation Conflict

Visual discomfort caused by mismatch between eye convergence (for 3D depth) and focus (for screen distance) in stereoscopic displays.

A company has released their new pair of auditory smart glasses, encompassing a front-mounted camera and directional microphone for sensing, and acoustically transparent audio output for user feedback. They have hired you to develop new applications for these glasses for two key use cases: accessibility and navigation. Propose two unique scenarios in everyday life where the smart glasses' combination of visual input and auditory output could benefit users with situational, temporary or permanent impairment. For each scenario, outline the impairment being addressed, the problem posed by the scenario or context, what pertinent contextual information the device could sense or infer, and what feedback could be given based on this to assist or support users.

The glasses could provide auditory navigation instructions for blind users who cannot navigate themselves and would not be able to notice hazards on their path. If these users are unaccompanied by a seeing person and do not know where they are going they could not easily find their way to a destination and are vulnerable to hazards in their path like bumps in the road or objects/people in the way. The glasses could use its cameras to find hazards for them and use its speakers to tell the user how to get around them or how to get to an unfamiliar destination, which directions to turn at what times for example. Additionally, if the user has to cross a road. The glasses could also be used by those who have difficulty hearing but are not necessarily deaf, for instance, to pick up audio around them and amplify it for them if the cameras see someone speaking to allow for the user to hear them better. In this scenario, if a bystander to the user is speaking to them, the camera would be able to detect them, and amplify their voice using the directional microphones while ignoring the ambient sounds around them.

Compare and contrast smartphone-based navigation (e.g. viewing Google Maps) with a prospective navigation app running on the company's smart glasses, reflecting on the benefits and limitations of smartphones vs smart glasses in this context.

First, smartphone-based navigation like google maps is heavily reliant on visual feedback, making this option not as accessible for visually impaired users who require auditory controls/feedback. Along similar lines, the smartphone-based navigation, if used for the same purpose of directing a blind user around obstacles to a destination, would require that the user holds their device in such a way that might look like they are pointing the cameras at other pedestrians, potentially impacting social acceptability - this is under the assumption that the smart glasses fit the form factor of standard glasses and so social acceptability would not be an issue. The smartphone would, however, be able to provide visual feedback through its display, for example displaying a map of the users route so they know what path to follow ahead of time. The smartphone approach diverts the users attention away from their route and to the device, where the glasses would not require the user to look away as they could continue to look at their path and listen for directions

2. A supermarket chain is creating an Augmented Reality (AR) shopping application for smartphones. The application will support users by efficiently guiding them around the store and helping users to locate items on their shopping lists on the shelves. A solution for delivering non-visual navigation feedback to guide users to products in the shop on their shopping lists is being proposed by your team. Identify the necessary wearable device(s) and output modalities used; reflect on how navigation feedback would be delivered; and suggest what sensing and contextual awareness would be required to enable this.

First, the user would wear a lightweight haptic belt (or wristband) containing four vibration motors arranged front-back and left-right, plus bone-conduction earphones. The belt delivers

directional haptic pulses—e.g. a single short buzz on the right signals “turn right,” a double buzz signals “you’ve arrived”—while the earphones provide spatialized audio beacons (a soft tone that seems to emanate from the aisle ahead) and brief text-to-speech prompts (“Coffee, aisle 3, just ahead on your left”). Next, once the user uploads their shopping list, the smartphone computes an optimised route through the store’s digital floor-plan. As the user walks, the belt vibrates on the side they need to turn about 2 m before each intersection; if they drift off the planned path, a low-frequency vibration “pings” from the correct direction. When they reach within 1 m of the target shelf, the earphones play a spatialized cue directly ahead, followed by a concise TTS label (“Oats, front shelf”). If the user taps the belt twice, the last instruction is repeated. Finally, accurate delivery depends on robust sensing and context: Bluetooth Low Energy beacons or UWB anchors spaced down each aisle provide 0.5 m positioning, smoothed by the phone’s IMU dead-reckoning when beacon signals fluctuate. The phone camera uses OCR to read end-of-aisle signs and shelf labels to confirm the correct zone, while the app’s inventory database flags out-of-stock items. The system also monitors walking speed (to adjust vibration timing), ambient noise (to raise or lower audio volume), and list progress (to dynamically re-optimize the remaining route).

A mobile app developer is designing a version of their popular fitness app for wrist-worn smartwatches, intended for use by joggers. The app provides feedback on athletic performance and controls music playback. The app developer wants to coach the user’s jogging performance based on their real-time physiological data and location tracking, to keep them at their desired pace and exertion. Using Augmented Reality (AR) glasses, propose both egocentric AND exocentric designs for delivering visual feedback to help users manage their jogging performance. Compare and contrast the safety of your designs.

In an egocentric design, AR content is locked to the user in a HUD style, for example showing the route via arrows on the ground, the time/distance ran or other metrics the joggers might need. This could obscure the users vision in some scenarios and create blindspots, which is a concern for safety

In the exocentric design, AR content is locked to the world, for example, in the style of signs or waypoint flags to convey the path and distance and again, present any content needed by the joggers. If AR content is misplaced, it could obscure physical objects or lead users the wrong direction.

You’ve been hired by the bank PJ Borgan to explore how XR technology could impact their business operations. Propose and justify two relevant use cases of XR, one each for VR and AR, and outline the potential benefits XR might offer in each case

VR could be leveraged for meetings within the company, as this can mimic the real experience of a face-to-face meeting even if those participating are not able to physically meet for any reason.

AR could be used to keep private information - which the company is likely to be working with as a bank - hidden. By displaying private information as AR content overlaid over the physical world, only the people who should have access to the information will be able to see it, removing the risk of others shoulder surfing to find out private information

Outline 4 potential risks posed by deploying XR headsets in banks for these use cases, and reflect on whether the technology is sufficiently mature that the bank should push for adoption.

In the AR for private information example, if the camera feeds required for the headsets are compromised, then a malicious person could get access to this and have easy access to the private information.

A potential risk deploying XR headsets, particularly over a pro-

longed period of time, could be vision problems like eye strain or cybersickness, both of which are common problems in XR

If we assume that the office is busy, having people in VR could pose a problem if the devices/software do not have proper reality or bystander awareness, as this could otherwise cause problematic interactions with surroundings

VR and AR devices, particularly for the use cases outlined, are likely to contain cameras to record their surroundings for proper functioning. This could be problematic to bystanders who do not know and/or do not wish to be recorded and for this to be collected. The company would likely need the explicit consent of everyone in the office to adopt this.

All things considered, the technology for this has matured enough to a degree that most devices have good systems for reality awareness, proper security, etc. and so this technology is sufficiently mature for the bank to adopt

Propose a navigation interaction technique to guide the user from their current position in store to the intended shelf location for their currently selected product. You should not describe or show a user interface layout. Instead, you should describe how users would interact with the system. You may wish to consider inputs to the system, any output or feedback it presents, and how it reacts to users' actions. Justify your design choices. You can propose the use of any interaction technologies covered during the course and can ignore the costs and feasibility of implementing your ideas.

First, the staff member activates the system hands-free by saying "Find SKU 12345," which highlights that product in the inventory database. A route is computed from their current BLE/SLAM position to the assigned shelf.

Next, haptic cues built into the temple pads of the glasses deliver left/right vibrations 2 m before each turn; a rapid double-pulse signals "you've arrived"

Simultaneously, spatialized audio beacons played over bone-conduction transducers guide the user down long aisles ("beep...beep" moving ahead), allowing eyes-up navigation without blocking ambient sounds.

If the user strays off-route, a low-frequency "ping" emanates from the correct direction, prompting course correction. Upon entering a 1 m radius of the target shelf, a brief TTS announcement—"Five boxes of granola here"—confirms the location, and a final haptic buzz auto-checks the item off their picking list

A passenger airline is developing a virtual reality training application for flight attendants. Users will be presented with a full-scale virtual aircraft occupied by virtual passengers, in which they will be guided through a series of operational and safety procedures. What locomotion technique would you recommend for this application? Justify your choice. You may wish to consider how your choice is affected by both the physical and virtual environments in which this system may be used.

First, because the virtual cabin is only a few metres deep and it's critical that attendants experience realistic sensorimotor cues, I recommend walking-in-place locomotion: the user physically steps on the spot, detected via head-and-foot IMUs, to move forward in VR. Head orientation dictates heading, and a gentle snap-turn button on the controller handles rotations.

This approach:

Maintains sensorimotor congruence, minimizing vection-induced cybersickness by matching visual flow to the user's own stepping motions.

Requires minimal physical space, since users never actually translate more than a metre in the real world.

Supports immersion and spatial memory, crucial for learning safety routes and service paths inside the aircraft.

Which of these approaches would you recommend? Justify your choice **Consider the alternative of using a mixed-reality display (e.g., augmented reality headset or a virtual-reality headset with video pass-through). This approach could be used on-board a real aircraft to present similar virtual content that guides users through safety and operational procedures.**

Which of these approaches would you recommend? Justify your choice First, for initial bulk training, I recommend the fully occlusive VR application in a studio: it offers complete control over passenger scenarios (e.g. turbulence, medical emergencies) and supports rich data capture (performance metrics, pause-and-rewind).

Next, for advanced on-aircraft drills, an AR pass-through headset (mixed reality) worn in a real cabin lets trainees practice in situ, overlaying virtual passengers and instructions on the genuine fuselage, seats, and galley. This in-context rehearsal solidifies muscle memory and spatial cues that VR alone cannot provide.